

# **Baligh**

Arabic Writing Assistant

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of

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## **Presented by**

**Ahmed Hamed Gaber Hamed**

**Akram Hany Karam Salam**

**Amir Anwar Bekhit Awd Kedis**

**Somia Saad Ismail El-Shemy**

## **Supervised by**

**Dr. Ayman AboElhassan**

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# Abstract

Arabic digital writing still suffers from a shortage of integrated tools that can correct Modern Standard Arabic text accurately, respond in real time, and explain their feedback in a way that helps users learn. Existing tools often focus on surface spelling fixes or black-box suggestions without exposing the grammatical reasoning behind the correction. Baligh addresses this problem by providing an Arabic writing assistant that combines grammatical error detection, grammatical error correction, spelling-aware editing, automatic vowelization (Tashkeel), next-word suggestion, and explanatory feedback in a single user-facing system. The project aims to support students, educators, and Arabic content writers by improving writing quality while also making the correction process more understandable and educational.

The implemented system follows a modular pipeline. A preprocessing layer performs normalization, word-boundary handling, segmentation, and morphological analysis and disambiguation. On top of that, Baligh integrates a grammatical error detection engine that fuses rule-based, lexicon-based, and machine-learning detectors, then passes its findings to correction components built around ontology-guided reasoning, dictionary lookup, text-editing models, and automatic vowelization support. In parallel, a next-word suggestion module supports both next-word prediction and word auto-completion. The final system highlights issues, proposes corrections, returns ranked suggestions, generates Tashkeel when needed, and presents grammar-oriented explanations to the user in an interactive workflow.

The project outputs include the preprocessing pipeline, grammatical error detection and correction services, the next-word suggestion subsystem, and a working user application. Testing and validation relied on automated unit and integration tests together with evaluation on established Arabic linguistic resources, including QALB14, QALB15, and ZAEBUC. Overall, Baligh demonstrates that a modular and explainable Arabic writing assistant can combine linguistic processing and machine learning into a practical platform for improving Arabic writing quality.

## الملخص

لا تزال الكتابة الرقمية باللغة العربية تعاني من نقص الأدوات المتكاملة التي تستطيع تصحيح نصوص العربية الفصحى بدقة، والعمل بزمان استجابة مناسب، وتقديم تفسير واضح يساعد المستخدم على التعلم بدل الاكتفاء بإظهار التصحيح فقط. فكثير من الأدوات المتاحة تركز على الأخطاء الإملائية السطحية أو تقدم اقتراحات غير مفسرة. يعالج مشروع بليغ هذه المشكلة من خلال بناء مساعد كتابة عربي يجمع بين اكتشاف الأخطاء النحوية، وتصحيحها، ومعالجة الأخطاء الإملائية، واقتراح الكلمة التالية، وتقديم تفسير مبسط للقاعدة أو سبب الخطأ داخل نظام واحد موجه للمستخدم. ويهدف المشروع إلى خدمة الطلاب والمعلمين وصناع المحتوى العربي عبر تحسين جودة الكتابة وجعل عملية التصحيح أكثر فائدة من الناحية التعليمية.

يعتمد النظام المنفذ على بنية معيارية تبدأ بطبقة للمعالجة المسبقة تشمل التطبيع، وتحديد حدود الكلمات، والتجزئة، ثم التحليل الصرفي وفك الالتباس. بعد ذلك يدمج بليغ محركاً لاكتشاف الأخطاء النحوية يعتمد على المزج بين قواعد لغوية، وكشافات معجمية، ونماذج تعلم آلي، ثم يمرر النتائج إلى وحدات التصحيح المبنية على الاستدلال المعتمد على الأنطولوجيا، والبحث المعجمي، ونماذج التحرير النصي. وبالتوازي مع ذلك، يوفر النظام وحدة لاقتراح الكلمة التالية تشمل التنبؤ بالكلمة التالية والإكمال التلقائي للكلمة الجارية كتابتها. وتعرض المنصة النهائية الأخطاء والتصحيحات والاقتراحات والتفسيرات اللغوية بصورة تفاعلية.

تشمل مخرجات المشروع خط معالجة مسبقة متكامل، وخدمات لاكتشاف الأخطاء وتصحيحها، ووحدة اقتراح الكلمات، وتطبيقاً عملياً للمستخدم. واعتمد التحقق من النظام على اختبارات آلية للوحدات والتكامل، إضافة إلى تقييم مبني على موارد عربية معروفة مثل QALB14 و QALB15 و ZAEUC. وتُظهر المحصلة أن بليغ يقدم نموذجاً عملياً لمساعد كتابة عربي قابل للتفسير يجمع بين المعالجة اللغوية والتعلم الآلي لخدمة الكتابة العربية بصورة أكثر جودة وفائدة.

# Acknowledgment

Baligh was not built through technical effort alone, but through the support, patience, and trust of many people around us. We would first like to thank our supervisor, Dr. Ayman AboElhassan, for his steady guidance throughout this project. His questions, feedback, and encouragement pushed us to think more carefully, work more seriously, and keep improving the project whenever we felt uncertain or stuck.

We are also deeply grateful to the Faculty of Engineering at Cairo University and the Department of Computer Engineering. The years we spent there shaped the way we think, learn, and solve problems, and this project reflects much of what we gained from that journey. Baligh became possible because of the academic foundation and sense of responsibility that this environment gave us.

We would also like to thank one another as a team. This project demanded time, patience, repeated trial and error, and many long discussions before ideas became clear and features became real. What made that process meaningful was not only the final result, but the way we supported each other through the difficult parts, shared the pressure, and kept moving forward together.

Finally, we thank our families and loved ones with all sincerity. Their patience during busy and stressful periods, their constant encouragement, and their belief in us gave us the strength to continue. Any achievement in this work carries a part of their support within it, and for that we are truly grateful.

# Contents

|                                                         |             |
|---------------------------------------------------------|-------------|
| <b>Abstract</b>                                         | <b>i</b>    |
| <b>المُلخَص</b>                                         | <b>ii</b>   |
| <b>Acknowledgment</b>                                   | <b>iii</b>  |
| <b>List of Figures</b>                                  | <b>v</b>    |
| <b>List of Tables</b>                                   | <b>vi</b>   |
| <b>List of Abbreviations</b>                            | <b>vii</b>  |
| <b>Contacts</b>                                         | <b>viii</b> |
| <b>1 Introduction</b>                                   | <b>1</b>    |
| 1.1 Motivation and Justification . . . . .              | 1           |
| 1.2 Project Objectives and Problem Definition . . . . . | 1           |
| 1.3 Project Outcomes . . . . .                          | 2           |
| 1.4 Document Organization . . . . .                     | 3           |
| <b>2 Market Visibility Study</b>                        | <b>4</b>    |
| 2.1 Targeted Customers . . . . .                        | 4           |
| 2.2 Market Survey . . . . .                             | 5           |
| 2.2.1 Qalam AI . . . . .                                | 5           |
| 2.2.2 Sahehly . . . . .                                 | 6           |
| 2.2.3 AlNahawi . . . . .                                | 7           |
| 2.2.4 Arwi . . . . .                                    | 7           |
| 2.3 Business Case and Financial Analysis . . . . .      | 8           |
| 2.3.1 Business Model . . . . .                          | 8           |
| 2.3.2 Pricing Assumptions . . . . .                     | 8           |
| 2.3.3 Capital Expenditure (CapEx) . . . . .             | 9           |
| 2.3.4 Operational Expenditure (OpEx) . . . . .          | 9           |
| 2.3.5 Revenue Projection . . . . .                      | 9           |
| 2.3.6 Cash Flow . . . . .                               | 9           |

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| 2.3.7    | Break-Even Analysis . . . . .                             | 10        |
| <b>3</b> | <b>Literature Survey</b>                                  | <b>11</b> |
| 3.1      | Background on Arabic NLP . . . . .                        | 11        |
| 3.2      | Background on Writing Assistance Systems . . . . .        | 11        |
| 3.3      | Comparative Study of Previous Work . . . . .              | 12        |
| 3.4      | Implemented Approach . . . . .                            | 13        |
| <b>4</b> | <b>System Design and Architecture</b>                     | <b>15</b> |
| 4.1      | Overview and Assumptions . . . . .                        | 15        |
| 4.2      | System Architecture . . . . .                             | 16        |
| 4.2.1    | Block Diagram . . . . .                                   | 17        |
| 4.3      | Module 1: Grammatical Error Detection . . . . .           | 17        |
| 4.3.1    | Functional Description . . . . .                          | 17        |
| 4.3.2    | GED Architecture . . . . .                                | 18        |
| 4.3.3    | Modular Decomposition . . . . .                           | 18        |
| 4.3.4    | Design Constraints . . . . .                              | 23        |
| 4.3.5    | Evaluation Summary . . . . .                              | 24        |
| 4.3.6    | GED Output in the GUI . . . . .                           | 25        |
| 4.4      | Module 2: Correction and Explanation Generation . . . . . | 25        |
| 4.4.1    | The Dictionary Module . . . . .                           | 25        |
| 4.4.2    | The Ontology Module . . . . .                             | 28        |
| 4.4.3    | Text Editing Sub-module . . . . .                         | 31        |
| 4.4.4    | Ranker Sub-module . . . . .                               | 36        |
| 4.5      | Module 3: Web Client . . . . .                            | 37        |
| 4.5.1    | Functional Description . . . . .                          | 37        |
| <b>5</b> | <b>System Testing and Verification</b>                    | <b>39</b> |
| 5.1      | Testing Setup . . . . .                                   | 39        |
| 5.2      | Testing Plan and Strategy . . . . .                       | 40        |
| 5.2.1    | Module Testing . . . . .                                  | 40        |
| 5.2.2    | Integration Testing . . . . .                             | 41        |
| 5.3      | Test Schedule . . . . .                                   | 42        |
| 5.4      | Comparative Results to Previous Work . . . . .            | 43        |
| 5.4.1    | Grammatical Error Detection (GED) Performance . . . . .   | 43        |
| 5.4.2    | Next-Word Prediction (NWP) Performance . . . . .          | 45        |
| 5.4.3    | Word Auto-Completion (WAC) Performance . . . . .          | 45        |
| <b>6</b> | <b>Conclusions and Future Work</b>                        | <b>46</b> |
| 6.1      | Faced Challenges . . . . .                                | 46        |

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| 6.1.1    | Research Challenges . . . . .                                     | 46        |
| 6.1.2    | Technical and Integration Challenges . . . . .                    | 46        |
| 6.2      | Gained Experience . . . . .                                       | 48        |
| 6.2.1    | Technical Skills . . . . .                                        | 48        |
| 6.2.2    | Soft Skills and Teamwork . . . . .                                | 48        |
| 6.3      | Conclusions . . . . .                                             | 49        |
| 6.4      | Future Work . . . . .                                             | 49        |
| 6.4.1    | Next-Word Suggestion (NWS) Enhancements . . . . .                 | 50        |
| 6.4.2    | Grammatical Error Detection and Correction Enhancements . . . . . | 50        |
| 6.4.3    | System and Platform Extensions . . . . .                          | 50        |
| <b>A</b> | <b>Development Platforms and Tools</b>                            | <b>53</b> |
| <b>B</b> | <b>Use Cases</b>                                                  | <b>56</b> |
| <b>C</b> | <b>User Guide</b>                                                 | <b>57</b> |
| <b>D</b> | <b>Code Documentation</b>                                         | <b>58</b> |
| <b>E</b> | <b>Feasibility Study</b>                                          | <b>59</b> |
| <b>F</b> | <b>GED Rules Reference</b>                                        | <b>60</b> |
| F.1      | Rule-Based GED Catalog . . . . .                                  | 60        |
| F.1.1    | Orthography Rules . . . . .                                       | 60        |
| F.1.2    | Punctuation Rules . . . . .                                       | 61        |
| F.1.3    | Syntax Rules . . . . .                                            | 61        |
| F.1.4    | Semantics Rules . . . . .                                         | 62        |
| F.2      | Lexicon Split and Merge Patterns . . . . .                        | 64        |
| F.2.1    | Split Patterns . . . . .                                          | 64        |
| F.2.2    | Merge Patterns . . . . .                                          | 65        |

# List of Figures

|      |                                                                                                                                                                         |    |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 2.1  | Wrong Correction . . . . .                                                                                                                                              | 6  |
| 2.2  | No Diacritiation Correction . . . . .                                                                                                                                   | 6  |
| 2.3  | Missed A Grammatical Mistake . . . . .                                                                                                                                  | 6  |
| 2.4  | Missed Many Grammatical Mistakes . . . . .                                                                                                                              | 6  |
| 2.5  | Wrong Correction . . . . .                                                                                                                                              | 6  |
| 2.6  | No Diacritization Correction . . . . .                                                                                                                                  | 6  |
| 2.7  | Missed The grammatical Error . . . . .                                                                                                                                  | 7  |
| 2.8  | Missed Many Grammatical Errors . . . . .                                                                                                                                | 7  |
| 2.9  | . . . . .                                                                                                                                                               | 7  |
| 2.10 | . . . . .                                                                                                                                                               | 7  |
| 2.11 | AlNahawi Interface (3) . . . . .                                                                                                                                        | 7  |
| 2.12 | . . . . .                                                                                                                                                               | 7  |
| 2.13 | Marked a correct word as a mistake and no Diacritics correction . . . . .                                                                                               | 8  |
| 3.1  | Example of text-editing tags for Arabic grammatical error correction. . . . .                                                                                           | 12 |
| 3.2  | Taxonomy of common Arabic grammatical and orthographic errors relevant to Baligh. . . . .                                                                               | 14 |
| 4.1  | High-level modular architecture of the Baligh system. . . . .                                                                                                           | 17 |
| 4.2  | GED detector architecture showing the rule-based, lexicon-based, and machine-learning detectors followed by the fusion and priority logic. . . . .                      | 18 |
| 4.3  | Fusion-layer flow used to resolve overlap conflicts and normalize final GED spans. . . . .                                                                              | 22 |
| 4.4  | Aggregate binary GED F1 score by subsystem from the evaluation results. . . . .                                                                                         | 24 |
| 4.5  | GED feedback inside the Baligh editor: highlighted detections in the writing view and an issue card showing explanation-oriented feedback for a detected error. . . . . | 25 |
| 4.6  | Spelling Error (Original) . . . . .                                                                                                                                     | 27 |
| 4.7  | Dictionary Correction . . . . .                                                                                                                                         | 27 |
| 4.8  | Typographical Error (Original) . . . . .                                                                                                                                | 27 |
| 4.9  | Dictionary Correction . . . . .                                                                                                                                         | 27 |
| 4.10 | Lexical Error (Original) . . . . .                                                                                                                                      | 27 |
| 4.11 | Dictionary Correction . . . . .                                                                                                                                         | 27 |



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|                                                                                                                                       |    |
|---------------------------------------------------------------------------------------------------------------------------------------|----|
| 4.12 Architecture of the GEC Module. . . . .                                                                                          | 29 |
| 4.13 Grammatical Error (Original) . . . . .                                                                                           | 30 |
| 4.14 Ontology Correction . . . . .                                                                                                    | 30 |
| 4.15 Syntactic Mismatch (Original) . . . . .                                                                                          | 30 |
| 4.16 Ontology Correction . . . . .                                                                                                    | 30 |
| 4.17 Case/Gender Error (Original) . . . . .                                                                                           | 31 |
| 4.18 Ontology Correction . . . . .                                                                                                    | 31 |
| 4.19 Tagger Pre-processing Pipeline . . . . .                                                                                         | 34 |
| 4.20 . . . . .                                                                                                                        | 35 |
| 4.21 . . . . .                                                                                                                        | 35 |
| 4.22 . . . . .                                                                                                                        | 35 |
| 4.23 . . . . .                                                                                                                        | 35 |
| 4.24 The Baligh web client: the public landing page and the editor workspace<br>used to present analysis results to the user. . . . . | 38 |

# List of Tables

|                                                                              |    |
|------------------------------------------------------------------------------|----|
| 2.1 Pricing Strategy . . . . .                                               | 9  |
| 3.1 Comparison of Related Arabic Writing Assistance Approaches . . . . .     | 13 |
| 4.1 Lexicon Resources Used by the GED Lexicon Detector . . . . .             | 20 |
| 4.2 ML Detector Selection Summary . . . . .                                  | 21 |
| 4.3 Implemented GED Subsystems and Their Roles . . . . .                     | 23 |
| 4.4 Aggregate Binary GED Performance Across All Evaluation Corpora . . . . . | 24 |
| 5.1 Grammatical Error Detection Subsystem Performance . . . . .              | 44 |
| 5.2 Next-Word Prediction (NWP) Accuracy Comparison . . . . .                 | 45 |
| 5.3 WAC Top-5 Accuracy by Typed Prefix Length . . . . .                      | 45 |

# List of Abbreviations

| Abbreviation | Meaning                      |
|--------------|------------------------------|
| AI           | Artificial Intelligence      |
| GEC          | Grammatical Error Correction |
| GED          | Grammatical Error Detection  |
| MSA          | Modern Standard Arabic       |
| NLP          | Natural Language Processing  |
| NWP          | Next-Word Prediction         |

# Contacts

## Team Members

| Name                        | Email                                                                                  | Phone Number |
|-----------------------------|----------------------------------------------------------------------------------------|--------------|
| Ahmed Hamed Gaber Hamed     | <a href="mailto:ahmed.hamed03@eng-st.cu.edu.eg">ahmed.hamed03@eng-st.cu.edu.eg</a>     | 01226968657  |
| Akram Hany Karam Salam      | <a href="mailto:akram.sallam03@eng-st.cu.edu.eg">akram.sallam03@eng-st.cu.edu.eg</a>   | 01148746563  |
| Amir Anwar Bekhit Awd Kedis | <a href="mailto:amir.awd03@eng-st.cu.edu.eg">amir.awd03@eng-st.cu.edu.eg</a>           | 01221461992  |
| Somia Saad Ismail El-Shemy  | <a href="mailto:somia.elshemy02@eng-st.cu.edu.eg">somia.elshemy02@eng-st.cu.edu.eg</a> | 01080732462  |

## Supervisor

| Name                  | Email                                                                                | Phone Number |
|-----------------------|--------------------------------------------------------------------------------------|--------------|
| Dr. Ayman AboElhassan | <a href="mailto:ayman.abo.elmaaty@eng.cu.edu.eg">ayman.abo.elmaaty@eng.cu.edu.eg</a> | 01060836189  |

# Chapter 1: Introduction

Baligh is an Arabic writing assistant designed to support Modern Standard Arabic writing through explainable correction and real-time assistance. This chapter introduces the motivation behind the project, defines the problem it addresses, summarizes the main outcomes delivered by the project, and outlines the organization of the remainder of the report.

## 1.1. Motivation and Justification

Arabic digital writing still lacks broadly available tools that combine grammatical support, writing speed, and educational value in a single user experience. While many writing assistants for other languages provide immediate correction, prediction, and revision support, Arabic users often rely on tools that address only isolated spelling mistakes, provide limited grammatical help, or return suggestions without clarifying the linguistic reason behind them. This gap is especially important in Modern Standard Arabic, where accurate writing is influenced by rich morphology, orthographic ambiguity, and context-sensitive grammatical structure.

These characteristics make Arabic NLP more demanding to model and more difficult to serve well in real time. A useful writing assistant must do more than detect surface-level errors; it must process text carefully, understand partially typed input, and return feedback quickly enough to support natural writing. It must also present corrections in a way that is understandable to users rather than functioning as a black box.

Baligh is motivated by this need for an integrated Arabic writing assistant that combines grammatical error detection, grammatical error correction, automatic vowelization (Tashkeel), next-word assistance, and explanatory feedback in one workflow. The project is intended to benefit students, educators, and Arabic content writers who need writing support that improves both text quality and user understanding at the same time.

## 1.2. Project Objectives and Problem Definition

The primary objective of Baligh is to build an explainable writing assistant for Modern Standard Arabic that can help users write more accurately and efficiently. To achieve this objective, the project integrates a preprocessing pipeline for normalization, seg-

mentation, and morphological analysis with a grammatical error detection subsystem, grammatical error correction components, an automatic vowelization (Tashkeel) capability, a next-word suggestion module, and a user-facing application interface. The system is designed to highlight issues, generate ranked suggestions, provide next-word and auto-completion support, generate vowelized output when needed, and present explanations that make the feedback more useful for learning.

In addition to functional assistance, the project aims to demonstrate that Arabic writing support can be delivered through a modular architecture that balances linguistic processing, machine learning, usability, and responsiveness. This makes Baligh not only a correction tool, but also a platform for combining multiple Arabic NLP capabilities into a practical workflow.

Formally, the problem addressed by this project is the lack of an integrated and explainable system for Modern Standard Arabic writing that can detect and correct grammatical issues, assist users during text entry, and deliver reliable feedback in a form suitable for real-time interactive use.

### 1.3. Project Outcomes

The project produced a working set of interconnected software components that together form the Baligh writing assistant. The implemented outcomes include a preprocessing pipeline responsible for text normalization, word-boundary handling, segmentation, and morphological analysis, as well as a grammatical error detection subsystem that combines rule-based, lexicon-based, and machine-learning detectors through a fusion layer.

The project also delivered grammatical error correction components that use ontology-based reasoning, dictionary-based lookup, and text-editing approaches to generate and rank correction candidates, together with automatic vowelization support for Arabic text. In parallel, Baligh includes a next-word suggestion subsystem that supports both next-word prediction and word auto-completion for interactive writing scenarios.

At the application level, the project provides a backend API and a web-based client that expose these capabilities to the user through an integrated writing workflow. Testing and evaluation were supported through automated checks and experiments on established Arabic resources, including QALB14, QALB15, and ZAEBUC, in addition to the accompanying project documents and technical materials.

## 1.4. Document Organization

The remainder of this report is organized as follows. Chapter 2 presents the Market Visibility Study, which examines the intended users of the system, reviews related products, and discusses the practical need for Baligh.

Chapter 3 provides the Literature Survey, covering the Arabic NLP background and the research context most relevant to the project. Chapter 4 describes the System Design and Architecture, including the main modules, their responsibilities, and the overall data flow.

Chapter 5 discusses System Testing and Verification, explaining how the implemented components were checked and evaluated. Finally, Chapter 6 presents the Conclusions and Future Work, summarizing the project contributions and outlining possible directions for extending Baligh in later stages.

## Chapter 2: Market Visibility Study

This chapter surveys the market related to the project and discusses similar tools or platforms. It should explain their limitations and justify the need for Baligh.

### 2.1. Targeted Customers

Baligh is designed to assist individuals and organisations that frequently produce Arabic text and require high grammatical accuracy and proper vowelisation. The system targets both educational and professional sectors, where maintaining linguistic correctness is essential.

The primary target customers include:

- **Students and Arabic Language Learners**

Students at schools and universities often struggle with Arabic grammar (Nahw) and the correct application of I'rab. Baligh provides real-time grammatical corrections and automatic vowelization.

- **Teachers and Educational Institutions**

Arabic language teachers can use Baligh as an educational assistant to review assignments, prepare teaching materials, and provide immediate feedback to students.

- **Researchers and Academic Writers**

Researchers who prepare scientific papers, theses, or publications in Arabic require grammatically correct and professionally written text.

- **Content Creators and Journalists**

Writers, journalists, bloggers, and digital content creators regularly produce large volumes of Arabic text. Baligh accelerates the proofreading process while maintaining linguistic accuracy.

- **Government and Business Organizations**

Government agencies and companies frequently produce official reports, legal documents, and correspondence in Arabic. Baligh reduces manual proofreading effort.

Customer Benefits:



- Real-time grammar correction based on Arabic Nahw and I'rab rules.
- Context-aware automatic vowelization (Tashkeel).
- Intelligent writing suggestions during typing.
- Reduced proofreading time.
- Improved readability and writing quality.
- Educational explanations that help users learn from their mistakes.

## 2.2. Market Survey

Although several Arabic writing assistants currently exist, none provides a comprehensive solution that combines grammatical reasoning, contextual vowelization, and real-time intelligent suggestions. Most existing systems specialize in only one aspect of Arabic language processing.

The following sections discuss the most relevant competing products.

### 2.2.1. Qalam AI

Qalam AI is an AI-powered Arabic writing assistant designed to improve spelling and grammar. It also supports automatic Tashkeel generation for Arabic text.

#### Strengths

- User-friendly interface.
- AI-assisted grammar and spelling correction.

#### Weaknesses

- Does not explain grammatical errors.
- Does not explicitly perform Nahw and I'rab analysis.
- Doesn't perform text completion.
- Doesn't correct diacritic mistakes.
- Not free.



Figure 2.1: Wrong Correction



Figure 2.2: No Diacritization Correction



Figure 2.3: Missed A Grammatical Mistake

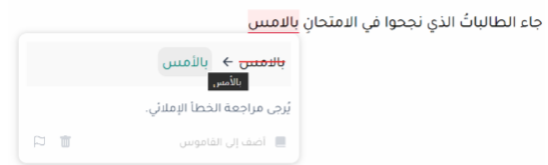


Figure 2.4: Missed Many Grammatical Mistakes

### 2.2.2. Sahehly

Sahehly is an Arabic proofreading platform that combines spelling correction, grammar checking, punctuation correction, and automatic Tashkeel.

#### Strengths

- Full-text diacritization.
- Doesn't detect diacritics mistakes.

#### Weaknesses

- Limited real-time interaction while typing.
- No explicit grammatical reasoning or I'rab analysis.
- Premium features require a subscription.

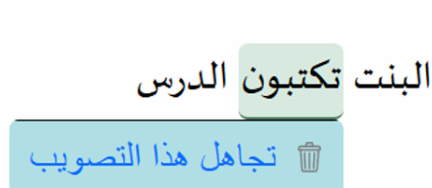


Figure 2.5: Wrong Correction

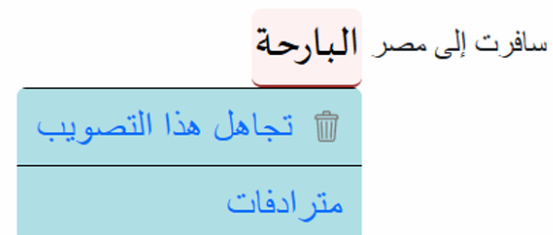


Figure 2.6: No Diacritization Correction



Figure 2.7: Missed The grammatical Error

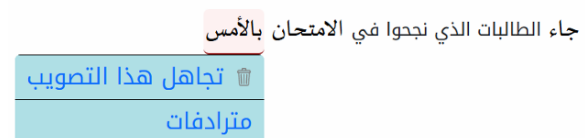


Figure 2.8: Missed Many Grammatical Errors

### 2.2.3. AlNahawi

Alnahawi is an AI-powered Arabic writing assistant designed to improve Arabic text through grammar correction and spelling correction.

#### Strengths

- Provides grammar and spelling correction.
- Simple and user-friendly interface.

#### Weaknesses

- Does not provide detailed grammatical explanations based on Nahw and I'rab.
- Doesn't perform text completion.
- Not fully free.

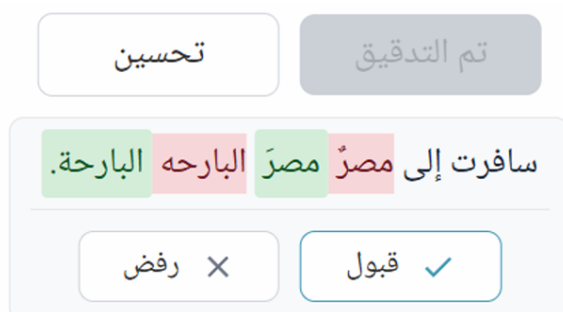


Figure 2.9

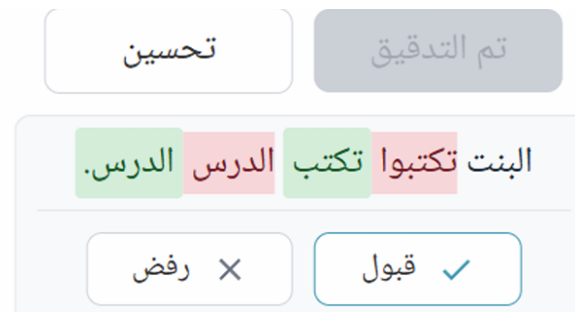


Figure 2.10

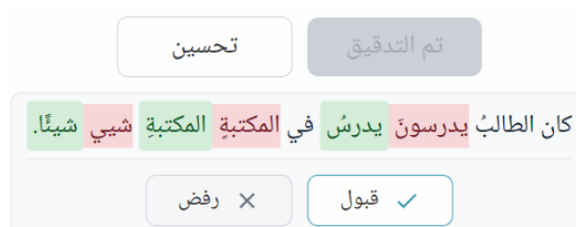


Figure 2.11: AlNahawi Interface (3)



Figure 2.12

### 2.2.4. Arwi

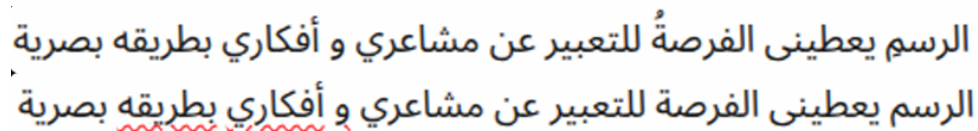
Arwi is an AI-based Arabic writing platform.

## Strengths

- Provides error detection.

## Weaknesses

- Doesn't perform error correction.
- Doesn't perform text completion.
- Doesn't correct diacritic mistakes.
- Does not provide detailed grammatical explanations based on Nahw and I'rab.
- Not free.



الرسم يعطيني الفرصة للتعبير عن مشاعري و أفكاري بطريقة بصرية  
الرسم يعطيني الفرصة للتعبير عن مشاعري و أفكاري بطريقة بصرية

**Figure 2.13:** Marked a correct word as a mistake and no Diacritics correction

## 2.3. Business Case and Financial Analysis

### 2.3.1. Business Model

Baligh can be offered using a Software-as-a-Service (SaaS) model with multiple subscription plans targeting individual users, educational institutions, and enterprise customers.

Potential revenue streams include:

- Individual monthly subscriptions.
- Educational institution licenses.
- Enterprise API access.

### 2.3.2. Pricing Assumptions

Pricing strategy is shown below.

**Table 2.1:** *Pricing Strategy*

| Plan       | Monthly Price    |
|------------|------------------|
| Free       | \$0              |
| Student    | \$5              |
| Individual | \$10             |
| Enterprise | Custom quotation |

### 2.3.3. Capital Expenditure (CapEx)

Estimated initial investment: **\$0**

### 2.3.4. Operational Expenditure (OpEx)

Monthly operating expenses include: **\$10 for API Hosting**

### 2.3.5. Revenue Projection

Assuming:

- 2,000 active users.
- 15% subscribe to the Student plan.
- 5% subscribe to the Individual plan.

Estimated monthly revenue:

- Student subscriptions:  $300 \times \$5 = \$1,500$
- Professional subscriptions:  $100 \times \$10 = \$1,000$

Total monthly revenue: **\$2,500**

Additional enterprise licensing and API subscriptions could significantly increase revenue.

### 2.3.6. Cash Flow

With a CapEx of \$0 and an OpEx of only \$10 per month, the project has a very low ongoing operating cost. As a result, the cash flow profile becomes highly favourable and even a small number of paying users would be sufficient to cover monthly expenses.

### **2.3.7. Break-Even Analysis**

The system is expected to reach break-even almost immediately after launch, and any subscription revenue beyond the monthly operating cost would generate positive cash flow from the start.

## Chapter 3: Literature Survey

This chapter summarizes the Arabic NLP background most relevant to Baligh and reviews the detection, correction and prediction approaches that influenced the system design.

### 3.1. Background on Arabic NLP

Arabic writing assistance is difficult because Arabic is morphologically rich, often undiacritized, and split across several language varieties [1, 2]. In Modern Standard Arabic, clitics attach to stems and words carry substantial grammatical information, so segmentation and morphological analysis are essential preprocessing steps. Tools such as Farasa and CAMEL Tools are therefore important foundations for downstream correction and prediction [1, 3].

Ambiguity is increased by the absence of short vowels in ordinary writing and by frequent orthographic issues such as Hamza variation, Ta-Marbuta confusion, punctuation mistakes, and merge or split errors [2, 4]. This means Arabic writing support must distinguish among orthographic, morphological, and syntactic errors rather than treat everything as spelling.

The main resources for Arabic GEC are still limited compared with English. QALB-2014 and QALB-2015 remain the primary benchmarks, while ZAEBUC and Tibyan extend coverage with additional corrected data and richer annotation [5–7]. Treebanks such as PADT, CAMEL Treebank, and I3rab are also useful because they expose grammatical structure that can support detection and explanation [8–10]. For next-word assistance, efficient language modeling remains important, with KenLM offering a practical low-latency option and neural models offering stronger context modeling at higher cost [11–13].

### 3.2. Background on Writing Assistance Systems

Writing assistants help users during text entry, not only after writing is complete. In Arabic, this usually requires a pipeline for normalization, segmentation, error detection, correction, ranking, and feedback presentation, with next-word assistance added when interactive typing support is needed.

Arabic GEC literature mainly falls into three families. Rule-based systems use hand-crafted rules, lexicons, and regular expressions, as in SAHSH@QALB-2015, and are attractive because they are interpretable [5]. Ontology-based systems, such as the approach of Moukrim et al., encode grammar more explicitly and support explanation-oriented correction [14]. Neural systems improve coverage and benchmark performance, especially with Transformer-based or pretrained seq2seq models [15,16]. Their main trade-off is lower interpretability and higher deployment cost.

Text-editing GEC is particularly important for Baligh. Instead of freely generating a corrected sentence, it predicts token-level edits, which makes the output faster and easier to align with user-visible changes. Alhafni and Habash showed that this approach is both accurate and efficient for Arabic [17].

|                               |                             |                         |                 |                            |                      |                            |                            |                   |               |                    |                   |                   |     |
|-------------------------------|-----------------------------|-------------------------|-----------------|----------------------------|----------------------|----------------------------|----------------------------|-------------------|---------------|--------------------|-------------------|-------------------|-----|
| Corrected                     | النفسية .<br><i>Alnfsyh</i> | الصححة<br><i>AlSHh</i>  |                 | سيما<br><i>symA</i>        | ولا<br><i>wLA</i>    | بالصححة<br><i>bAlSHh</i>   | الاهتمام<br><i>AlAhtmA</i> | يجب<br><i>yjb</i> | (a)           |                    |                   |                   |     |
|                               | 7                           | 6                       | 5               | 4                          | 3                    | 2                          | 1                          | 0                 |               |                    |                   |                   |     |
| Erroneous                     | النفسية<br><i>Alnfsyh</i>   | الصححة<br><i>AlSHh</i>  | في<br><i>fy</i> | ولاسيما<br><i>wLA symA</i> | لصححة<br><i>lSHh</i> | ب<br><i>b</i>              | الإهتمام<br><i>AlĀhtmA</i> | يجب<br><i>yjb</i> | (b)           |                    |                   |                   |     |
| Word Edits                    | KKKKKKR_[:̣]A_[.]           | KKKKR_[:̣]              | DD              | KKKI_[ ]KKKK               | MI_[']KKKR_[:̣]      | K                          | KKR_[']KKKKK               | KKK               | (c)           |                    |                   |                   |     |
| Word Edits<br>(Compressed)    | K*R_[:̣]A_[.]               | K*R_[:̣]                | D*              | KKKI_[ ]K*                 | MI_[']K*R_[:̣]       | K*                         | KKR_[']K*                  | K*                | (d)           |                    |                   |                   |     |
|                               | 7b                          | 7a                      | 6b              | 6a                         | 5                    | 4                          | 3b                         | 3a                | 2             | 1b                 | 1a                | 0                 |     |
| Tokenized<br>Erroneous        | ##ه<br>##h                  | النفسى<br><i>Alnfsy</i> | ##ه<br>##h      | الصح<br><i>AlSH</i>        | في<br><i>fy</i>      | ولاسيما<br><i>wLA symA</i> | ##ه<br>##h                 | لصح<br><i>lSH</i> | ب<br><i>b</i> | ##هتتام<br>##hhtmA | الإ<br><i>AlĀ</i> | يجب<br><i>yjb</i> | (e) |
| Subword Edits                 | R_[:̣]A_[.]                 | KKKKKK                  | R_[:̣]          | KKKK                       | DD                   | KKKI_[ ]KKKK               | R_[:̣]                     | MI_[']KKK         | K             | KKKKK              | KKR_[']           | KKK               | (f) |
| Subword Edits<br>(Compressed) | R_[:̣]A_[.]                 | K*                      | R_[:̣]          | K*                         | D*                   | KKKI_[ ]K*                 | R_[:̣]                     | MI_[']K*          | K*            | K*                 | K*R_[']           | K*                | (g) |

Figure 3.1: Example of text-editing tags for Arabic grammatical error correction.

Writing assistants also need grammatical error detection and useful feedback. Treating GED as a sequence labeling task can improve interpretability, and resources such as ARETA help classify correction outputs into meaningful error types [4, 16]. For next-word assistance, n-gram models remain attractive for low latency, while neural and character-aware models offer stronger contextual modeling when more computation is acceptable [11, 12, 18]. ARWI is a strong example of a more complete Arabic writing assistant built around these ideas [19].

### 3.3. Comparative Study of Previous Work

Previous work shows a clear trade-off. Symbolic systems are more explainable, but neural systems usually provide broader coverage and better benchmark results. Text-editing models offer a useful compromise because they keep the correction output structured while remaining much faster than fully generative systems [5, 14, 15, 17].



The literature also shows that strong Arabic writing support depends on more than one model. Preprocessing tools such as Farasa and CAMEL Tools are needed before correction, benchmark datasets shape what systems can learn, and predictive text models address a separate but complementary part of the user experience [1, 3, 11]. The main gap is therefore integration: many papers solve one subproblem well, but fewer systems combine correction, prediction, and explanation in one practical workflow.

**Table 3.1:** Comparison of Related Arabic Writing Assistance Approaches

| Work/System             | Main Approach                         | Strengths                      | Limitations                                |
|-------------------------|---------------------------------------|--------------------------------|--------------------------------------------|
| SAHSOH@QALB-2015 [5]    | Rule-based correction on QALB         | Explainable and lightweight    | Limited coverage                           |
| Moukrim et al. [14]     | Ontology-based syntactic correction   | Strong grammar grounding       | Hard to scale                              |
| Solyman et al. [15]     | Transformer-based Arabic GEC          | Strong benchmark results       | Heavier and less interpretable             |
| Alhafni et al. [16]     | Pretrained seq2seq GEC with GED input | State-of-the-art performance   | Model-heavy pipeline                       |
| Alhafni and Habash [17] | Text-editing GEC                      | Fast and accurate              | Needs integration with full product layers |
| ARWI [19]               | Arabic writing assistant workflow     | Closest system-level reference | Learner-oriented scope                     |

### 3.4. Implemented Approach

The survey suggests that Baligh should not depend on a single method. A purely rule-based system is easy to explain but limited in coverage, while a purely neural system is stronger statistically but harder to interpret and deploy. Baligh therefore adopts a hybrid design that combines linguistic preprocessing, explainable correction logic, and lightweight predictive assistance.

At the front of the pipeline, Baligh uses morphology-aware preprocessing because segmentation, normalization, and disambiguation are essential for Arabic NLP [1, 3]. For correction, the literature supports a layered strategy: symbolic rules and ontology-inspired ideas remain useful for precision and explanation, while modern GEC findings, especially text-editing approaches, offer better scalability and speed [5, 14, 17].

| Class       | Err Tag | Description                            | Arabic Example              | Transliteration            |
|-------------|---------|----------------------------------------|-----------------------------|----------------------------|
| Orthography | OA      | Confusion in Alif, Ya and Alif-Maqsurā | علي ← على                   | çly → çlȳ                  |
|             | OC      | Wrong order of word characters         | تيرينا ← ترينا              | tbrynA → trbynA            |
|             | OD      | Additional character(s)                | يعنوم ← يدوم                | yçdwm → ydwm               |
|             | OG      | Lengthening short vowels               | نقيمو ← نقيم                | nqymw → nqym               |
|             | OH      | Hamza errors                           | اكثر ← أكثر                 | Akθr → Âkθr                |
|             | OM      | Missing character(s)                   | سائلين ← سائلين             | sAlȳn → sAȳȳlyn            |
|             | ON      | Confusion between Nun and Tanwin       | ثوين ← ثوب                  | θwbñ → θwbū                |
|             | OR      | Replacement in word character(s)       | مصلنا ← وصلنا               | mSlnA → wSlnA              |
|             | OS      | Shortening long vowels                 | أوقت ← أوقات                | Âwqt → ÂwqAt               |
|             | OT      | Confusion in Ha, Ta and Ta-Marbuta     | مشاركة ← مشاركة             | mšArkħ → mšArkħ            |
|             | OW      | Confusion in Alif Fariqa               | وكانو ← وكانوا              | wkAnw → wkAnwA             |
|             | OO      | Other orthographic errors              | -                           | -                          |
| Morphology  | MI      | Word inflection                        | معروف ← عارف                | mçrwf → çArf               |
|             | MT      | Verb tense                             | تفرحتني ← أفرحتني           | tfrHny → ÂfrHtny           |
|             | MO      | Other morphological errors             | -                           | -                          |
| Syntax      | XC      | Case                                   | رائع ← رائعاً               | rAȳç → rAȳçAā              |
|             | XF      | Definiteness                           | السن ← سن                   | Alsn → sn                  |
|             | XG      | Gender                                 | العربي ← العربية            | Alȳrby → Alȳrbyh           |
|             | XM      | Missing word                           | على ← Null                  | Null → çly                 |
|             | XN      | Number                                 | فكرتي ← أفكارتي             | fkrty → ÂfkAry             |
|             | XT      | Unnecessary word                       | على ← Null                  | çly → Null                 |
|             | XO      | Other syntactic errors                 | -                           | -                          |
| Semantics   | SF      | Conjunction error                      | سبحان ← فسبحان              | sbHAN → fsbHAN             |
|             | SW      | Word selection error                   | من ← عن                     | mn → çn                    |
|             | SO      | Other semantic errors                  | -                           | -                          |
| Punctuation | PC      | Punctuation confusion                  | المتوسط ← المتوسط           | AlmtwsT. → AlmtwsT,        |
|             | PM      | Missing punctuation                    | العظيم ← العظيم،            | AlçDȳm → AlçDȳm,           |
|             | PT      | Unnecessary punctuation                | العام ← العام،              | AlçAm, → AlçAm             |
|             | PO      | Other errors in punctuation            | -                           | -                          |
| Merge       | MG      | Words are merged                       | ذهبت البارحة ← ذهبت البارحة | ðhbtAlbArHh → ðhbt AlbArHh |
| Split       | SP      | Words are split                        | المحادثات ← المحادثات       | AlmHA dθAt → AlmHADθAt     |

Table 1: The ALC error type taxonomy extended with merge and split classes. The error tags are listed alphabetically, except for the highlighted *\*Other* tags, which we do not support in ARETA.

**Figure 3.2:** Taxonomy of common Arabic grammatical and orthographic errors relevant to Baligh.

The same reasoning applies to prediction. Since Baligh is interactive, efficient language modeling is a practical baseline for next-word suggestion, with neural models reserved for cases where extra context modeling justifies the cost [11, 12]. Separating detection, correction, and explanation is also consistent with the literature because it improves interpretability and makes user feedback more informative [4, 16].

In summary, Baligh follows the literature by combining morphology-aware preprocessing, structured error handling, explainable correction support, and efficient prediction into one real-time writing workflow.

# Chapter 4: System Design and Architecture

This chapter forms the core of the report, detailing the technical foundation of Baligh. It outlines the design principles and structural components that enable the system to perform high-accuracy Grammatical Error Correction (GEC) and Next-Word Suggestion (NWS) with real-time responsiveness. Beginning with the high-level system architecture and then diving deep into the specific inner workings of each module and submodule.

## 4.1. Overview and Assumptions

Baligh is designed as a comprehensive Modern Standard Arabic (MSA) writing assistant that aims to provide real-time Grammatical Error Correction (GEC) and Next-Word Suggestion (NWS). The primary goal is to achieve high linguistic accuracy combined with low latency to ensure a seamless typing experience. To achieve this, the system is architected to differentiate between slow-path operations (such as deep grammatical checking) and fast-path operations (such as real-time predictive text).

During the design and implementation of the system, several key assumptions and technical constraints were established:

- **Joint Morphological Processing:** Diacritization and morphological analysis are mutually dependent in Arabic. The system assumes that morphological disambiguation implicitly provides the correct diacritization based on context, meaning they are resolved in a single joint pass rather than sequentially.
- **Real-Time Word Boundary Handling:** Since user input is instantaneous and often contains incomplete words at the cursor, the system assumes that only tokens terminated by specific delimiters (whitespace or punctuation) are fully formed. Incomplete fragments bypass the full morphological pipeline and are directly routed to the word auto-completion engine.
- **Modularity for Explainability:** The system assumes that users need understandable feedback. Thus, error detection and correction are loosely coupled, allowing the Explanation/Fusion layer to compile clear reasoning from multiple detection subsystems (rule-based, pattern-based, and ML).

## 4.2. System Architecture

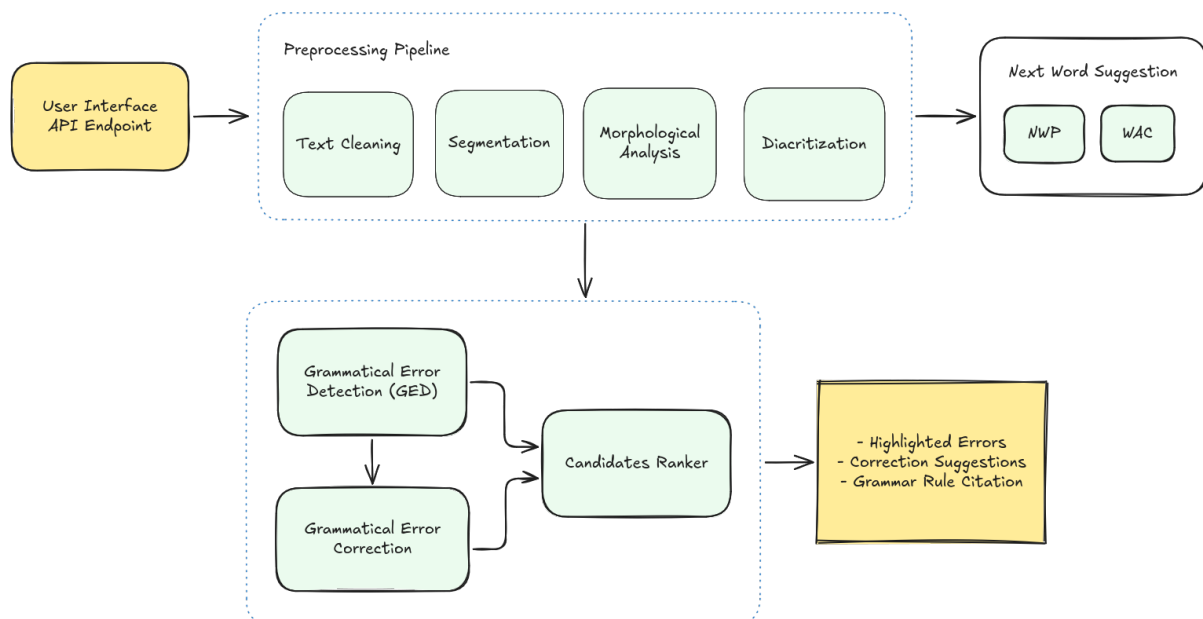
The system architecture of Baligh is highly modular, orchestrating data through a carefully constructed pipeline that balances deep linguistic analysis with immediate user responsiveness. The architecture is broadly separated into three main layers: the Client Layer, the Preprocessing Pipeline, and the Core Features Layer.

Data enters the system through the **Client Layer**, which consists of the user interface (web text editor) and the REST API backend. The raw text is immediately sent to the **Preprocessing Pipeline**. This pipeline performs foundational text cleaning, normalization, tokenization via Farasa, and contextual morphological analysis and disambiguation using CAMEL tools.

Once the text is preprocessed, it is dispatched into the **Core Features Layer**, which splits into two parallel tracks:

1. **The Slow Path (GEC & GED):** Preprocessed text is fed into the Grammatical Error Detection (GED) Engine, which utilizes a fusion of rule-based checkers, pattern matchers, and machine-learning sequence labelers to identify errors. These findings are passed to the Grammatical Error Correction (GEC) Module, which proposes corrections via an ontology-guided grammar lookup, dictionary spelling checks, and a text-editing machine learning model. Each module generates human-readable explanations. A final fusion and ranking layer consolidates the corrections and explanations, ensuring that the user receives the most relevant and understandable feedback.
2. **The Fast Path (NWS):** For predictive typing, the Next-Word Suggestion (NWS) module operates with minimal latency. It relies on a smart caching layer (for idioms and user patterns) and a context-aware trie-based lookup for word auto-completion (WAC) or next-word prediction (NWP).

### 4.2.1. Block Diagram



**Figure 4.1:** High-level modular architecture of the Baligh system.

## 4.3. Module 1: Grammatical Error Detection

The Grammatical Error Detection (GED) module is the primary analysis engine responsible for identifying linguistic issues in preprocessed Arabic text. It receives normalized tokens and morphological features from the preprocessing pipeline and produces a unified set of annotated error spans, each enriched with a category label, confidence score, and optional explanation. The GED module follows a multi-detector architecture in which three complementary subsystems run in parallel before a fusion layer reconciles their outputs into one consistent result set.

### 4.3.1. Functional Description

The GED module operates on the output of the preprocessing pipeline, which supplies segmented tokens, part-of-speech tags, morphological features, and disambiguation results. Its primary function is to identify spans of text that contain grammatical, orthographic, punctuation, or semantic errors and to assign each span an error category from the Baligh tag schema. Because the module is designed for an educational writing assistant, it also attaches human-readable explanations to rule-derived findings so that users can understand the reasoning behind each detection.

The module is invoked as part of the slow path in the Baligh architecture, meaning it runs after the user submits text for analysis rather than during keystroke-level interaction.

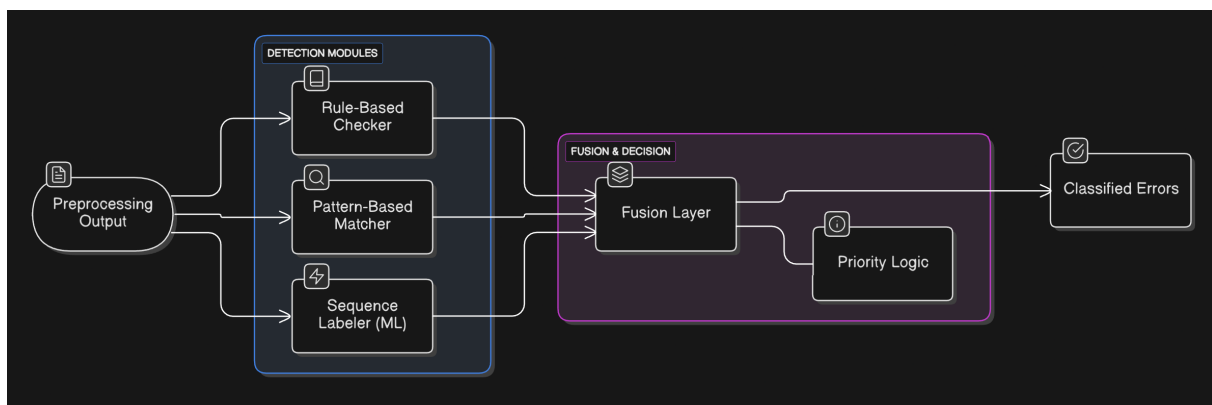
This design allows the detectors to perform deeper linguistic checks without impacting the responsiveness of the fast-path next-word suggestion system.

### 4.3.2. GED Architecture

The GED subsystem is built around a layered detector architecture that combines symbolic and statistical approaches. Rather than relying on a single model family, the design separates concerns across three detector lanes and one fusion stage:

1. **Rule-Based Detector:** applies explicit linguistic checks using handcrafted rules.
2. **Lexicon Matcher:** detects curated lexical patterns, spelling anomalies, and split/merge errors using trie-backed resources.
3. **Machine-Learning Sequence Labeler:** predicts token-level GED labels using a trained CRF model with morphological features.
4. **Fusion Layer:** reconciles overlapping outputs from all three detectors into one final error set.

This hybrid strategy was motivated by the literature, which showed that symbolic systems remain useful in Arabic when interpretability is important, while learned approaches improve coverage on broader contexts [5, 14, 16].



**Figure 4.2:** GED detector architecture showing the rule-based, lexicon-based, and machine-learning detectors followed by the fusion and priority logic.

### 4.3.3. Modular Decomposition

#### Rule-Based Subsystem

The rule-based detector is the most interpretable lane in the GED stack. It applies explicit linguistic checks to preprocessed text and produces high-confidence, explanation-

friendly findings that can be verified deterministically. The subsystem supports a hybrid rule-hosting model:

- **Declarative YAML rules:** compact pattern-driven checks defined in structured configuration files, well-suited for token-matching and simple morphological conditions.
- **Procedural Python rules:** richer checks that require multi-token context, complex conditionals, or access to morphological feature combinations that cannot be expressed declaratively.

Both rule types are routed through a common registry that handles loading, validation, and execution. The rule authoring workflow involved sourcing grammar material from LanguageTool and online Arabic grammar references, drafting rules in the internal format, manually reviewing each rule against the tag schema and preprocessing assumptions, and testing them individually before acceptance.

The implemented rule base covers four categories: orthography rules (Hamza placement, Alif Maqsura, Ta Marbuta, Idgham patterns), punctuation rules (spacing and Latin-to-Arabic character variants), syntax rules (demonstrative-noun agreement, jussive operators, verb-subject agreement, preposition case requirements), and semantics rules (lexical usage preferences and stylistic recommendations). The full catalog is documented in Appendix F.

## Lexicon Subsystem

The lexicon subsystem extends detection beyond procedural rules by combining curated token and split/merge patterns with trie-backed lookup and conservative spelling suspicion. Its role is to act as a middle layer between rigid symbolic rules and broader machine-learned predictions.

- **General word lexicon:** used to determine whether a normalized token is a plausible in-vocabulary Arabic word. Contains 9,009,550 entries.
- **Named-entity phrase lexicon:** used to protect people, places, organizations, and other entity mentions that may not appear in the general word list. Contains 60,833 phrases.
- **Named-entity token lexicon:** provides token-level entity coverage with 43,292 entries.

The word lexicon supports ordinary spelling lookup, while the entity lexicons reduce false positives on proper nouns and multi-token names. Additionally, the lexicon de-

lector maintains curated split patterns (e.g., detecting when **بعد ما** should be written as **بعدهما**) and merge patterns (e.g., detecting when **انشاء الله** should be split into **الله** **إن شاء الله**).

**Table 4.1:** *Lexicon Resources Used by the GED Lexicon Detector*

| Resource                    | Size              |
|-----------------------------|-------------------|
| General word lexicon        | 9,009,550 entries |
| Named-entity phrase lexicon | 60,833 phrases    |
| Named-entity token lexicon  | 43,292 tokens     |

## Machine-Learning Subsystem

The machine-learning subsystem treats GED as a token-level sequence-labeling task over the Baligh error categories. The final deployed model is a morphology-aware Conditional Random Field (CRF) trained on QALB14 and QALB15 data, covering 19,706 sentences and 1,063,125 tokens.

**Feature engineering.** The CRF model uses a rich feature set organized into four groups:

- **Surface features:** normalized token form, token shape, token length, Arabic / digit / punctuation flags, prefixes and suffixes up to length four, sentence-boundary markers, neighboring normalized tokens, sentence-length and position buckets.
- **Orthographic cues:** diacritic presence, repeated-character indicators, definite-article cues, and character n-grams.
- **Morphology features:** POS, UD POS, lexeme, stem, root, pattern, gender, number, person, aspect, voice, mood, state, case, proclitic and enclitic fields.
- **Confidence and context features:** backoff-source indicators, discretized morphology log-probability buckets, and neighboring POS features.

**Model selection.** Multiple model architectures were evaluated during development, including token-memory heuristics, calibrated token memory, linear classifiers, and CRF sequence models. The final selected model was the morphology-enhanced CRF (`crf_surface_v1_morph_v1`), which delivered the strongest development performance.



**Table 4.2:** *ML Detector Selection Summary*

| Role              | Model            | Threshold | Precision | Recall | F1     |
|-------------------|------------------|-----------|-----------|--------|--------|
| Earlier candidate | Surface-only CRF | 0.35      | 0.9080    | 0.8070 | 0.8545 |
| Final deployed    | CRF + morphology | 0.40      | 0.9158    | 0.8332 | 0.8725 |

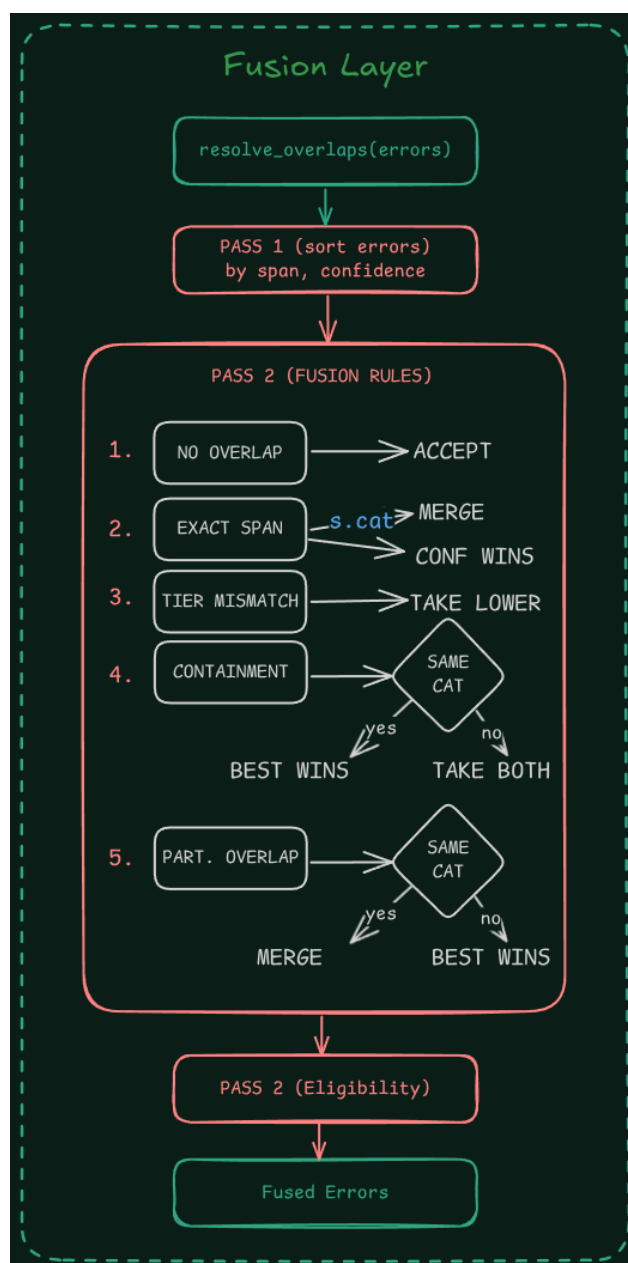
The final runtime detector is loaded from a model bundle with an inference manifest that keeps the training-time feature contract aligned with deployment.

### Fusion Layer

The fusion layer is essential because a multi-detector system is only useful if overlapping outputs can be reconciled consistently. The implemented algorithm performs the following steps:

1. Sort all detected spans by start position and length.
2. Perform a single left-to-right sweep over the accepted list.
3. Apply resolution rules for exact-span duplicates, containment, partial overlap, and tier mismatches.
4. Normalize explanation eligibility so that statistical predictions do not claim rule-like explanations they do not possess.

The tier system assigns priority levels: Tier 1 rule-derived detections have the highest confidence and can override statistical detections on the same span, while Tier 3 statistical detections provide broader coverage for spans not addressed by symbolic detectors.



**Figure 4.3:** Fusion-layer flow used to resolve overlap conflicts and normalize final GED spans.

**Table 4.3:** *Implemented GED Subsystems and Their Roles*

| Subsystem           | Purpose                                             | Main Strength                                   | Implementation Approach                               |
|---------------------|-----------------------------------------------------|-------------------------------------------------|-------------------------------------------------------|
| Rule-based detector | Apply linguistic checks over pre-processed text     | Precision and explainability                    | Hybrid YAM-L/Python rule hosting with registry        |
| Lexicon matcher     | Detect curated lexical and spelling-oriented issues | Bridge between rules and statistical prediction | Trie-backed resources with pattern matching           |
| Sequence labeler    | Predict token-level GED labels from features        | Broad contextual coverage                       | CRF with morphology-aware feature engineering         |
| Fusion layer        | Merge subsystem outputs into one final result set   | Stable multi-detector behavior                  | Tier-based overlap resolution with explanation gating |

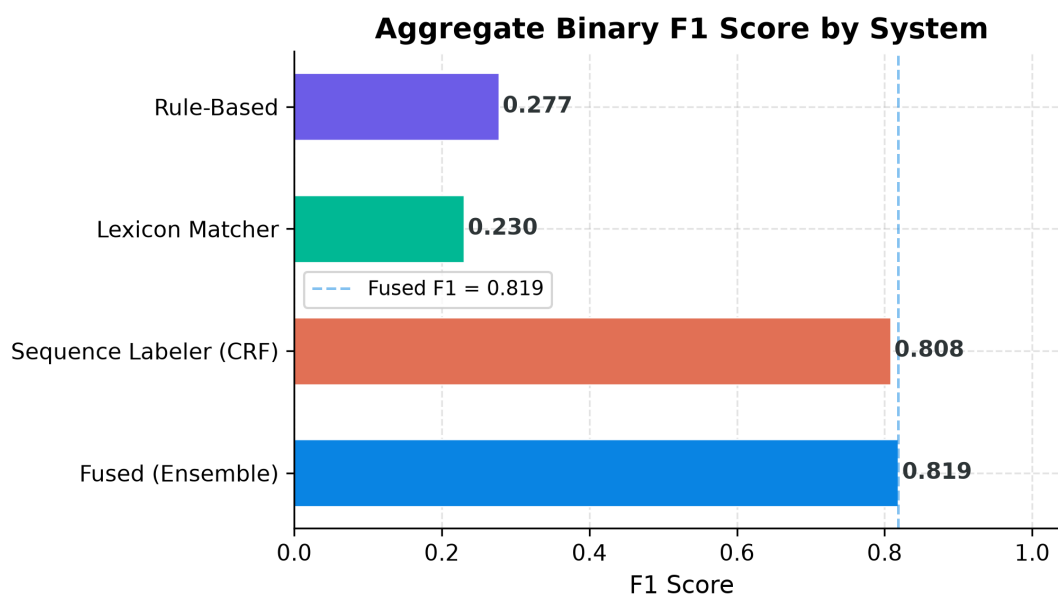
#### 4.3.4. Design Constraints

Several constraints shaped the design of the GED module:

- **Arabic morphological ambiguity:** Arabic tokens are highly ambiguous due to the absence of short vowels in ordinary writing, clitic attachment, and context-dependent grammatical role assignment. The GED module depends on reliable morphological disambiguation from the preprocessing pipeline to produce meaningful detections.
- **Explainability requirements:** Because Baligh targets educational use, GED outputs must carry human-readable explanations rather than opaque labels. This constraint motivated the inclusion of rule-based and lexicon-based detectors alongside the statistical component.
- **Data scarcity:** High-quality labeled Arabic GED data is limited compared with English resources. The CRF model was trained on QALB14 and QALB15, and the rule base was developed to extend coverage to error types underrepresented in training data.
- **Latency tolerance:** The GED module operates on the slow path, meaning it runs after the user submits text rather than during typing. This allows deeper analysis but requires that the combined detector and fusion pipeline completes within a reasonable response window.

### 4.3.5. Evaluation Summary

The GED module was evaluated across multiple corpora including QALB14, QALB15 (L1 and L2), and ZAEBUC. The full per-corpus breakdown is presented in Chapter 5 (Table 5.1). The aggregate binary results demonstrate that the symbolic lanes contribute precise but narrower coverage, while the learned and fused systems deliver stronger end-to-end performance.



**Figure 4.4:** Aggregate binary GED F1 score by subsystem from the evaluation results.

**Table 4.4:** Aggregate Binary GED Performance Across All Evaluation Corpora

| System           | Precision | Recall | F1    |
|------------------|-----------|--------|-------|
| Rule-Based       | 0.685     | 0.174  | 0.277 |
| Lexicon Matcher  | 0.892     | 0.132  | 0.230 |
| Sequence Labeler | 0.913     | 0.725  | 0.808 |
| Fused            | 0.905     | 0.747  | 0.819 |

These numbers should be interpreted with care because the error categories are not evenly distributed across the evaluation datasets. In practice, the datasets highlight models that perform well on the more frequent tags, which naturally favors the sequence labeler. However, the rule-based and lexicon-based lanes still detect error types that the sequence labeler may miss, especially categories that are less common in the training data. The fused gain demonstrates that symbolic detectors contribute useful coverage beyond the dominant dataset patterns.

### 4.3.6. GED Output in the GUI

Because Baligh is ultimately a user-facing writing assistant, the detector output must be understandable in the editor rather than only measurable in an evaluation script. Figure 4.5 shows two real GUI captures: one emphasizing highlighted detections inside the editor text, and one showing the issue card and explanation-oriented presentation attached to a detected span.



**Figure 4.5:** GED feedback inside the Baligh editor: highlighted detections in the writing view and an issue card showing explanation-oriented feedback for a detected error.

## 4.4. Module 2: Correction and Explanation Generation

The Grammatical Error Correction (GEC) module processes text to generate accurate correction candidates for grammatical and spelling mistakes. This module is divided into three primary components: the Dictionary Module, the Ontology Module, and the Edit Tagger Module.

### 4.4.1. The Dictionary Module

#### Functional Description

The Dictionary Module is responsible for robust spelling correction, Out-Of-Vocabulary (OOV) detection, and serving as a morphological generation backbone for the ontology. The Arabic language presents unique challenges in lexical validation due to its

highly inflectional nature. To address this, the module utilizes the **Arramooz** dictionary, an open-source comprehensive Arabic lexicon organized in relational database tables. Arramooz encapsulates approximately 6,000 roots and covers stop words, nouns, and verbs.

The complete dictionary pipeline involves detecting OOV words by stripping clitics (prefixes/suffixes) to isolate the stem, querying the database for close string matches, and retrieving spelling candidates.

## Modular Decomposition

The Dictionary Engine consists of the following submodules:

- **Spelling Checker:** Analyzes the incoming tokens and performs stem-based clitic analysis. Then it detects Out-Of-Vocabulary (OOV) words and generates a list of spelling candidates using edit distance metrics.
- **Arramooz Client:** A wrapper designed for efficient, low-latency querying of the SQLite-based Arramooz database.
- **Morphological Generator:** Acts upon requests from the Ontology Engine. When a specific inflection is required (e.g., converting a singular noun to a plural form in the accusative case), this generator queries Arramooz to construct the morphologically correct word form.
- **Alternative Ranker:** Ranks spelling candidates using Levenshtein distance (edit distance) combined with corpus-frequency metrics to present the most probable correction first.

## Design Constraints

The primary constraint of the Dictionary Module is the reliance on raw string similarity (edit distance) for initial spelling suggestions. Edit distance lacks context-awareness. To counter this, the Dictionary Engine relies on downstream contextual rankers (like the Machine Learning fusion layer) to make the final decision among its proposed alternatives.

## Results

قد تم التوصل لهذا الاستنتاج

Figure 4.6: Spelling Error (Original)



Figure 4.7: Dictionary Correction

يتم تحديث البيانات

Figure 4.8: Typographical Error (Original)



Figure 4.9: Dictionary Correction

ذهبت الامس إلى النادي

Figure 4.10: Lexical Error (Original)



Figure 4.11: Dictionary Correction

## 4.4.2. The Ontology Module

### Functional Description

An **Ontology** in computer science is a formal, explicit specification of a shared conceptualization. It defines a set of concepts and categories in a subject area, showing their properties and the relations between them. In the context of Baligh, the Ontology Module encodes Arabic syntax rules into a computational format, inspired by dependency grammar based on predicate logic.

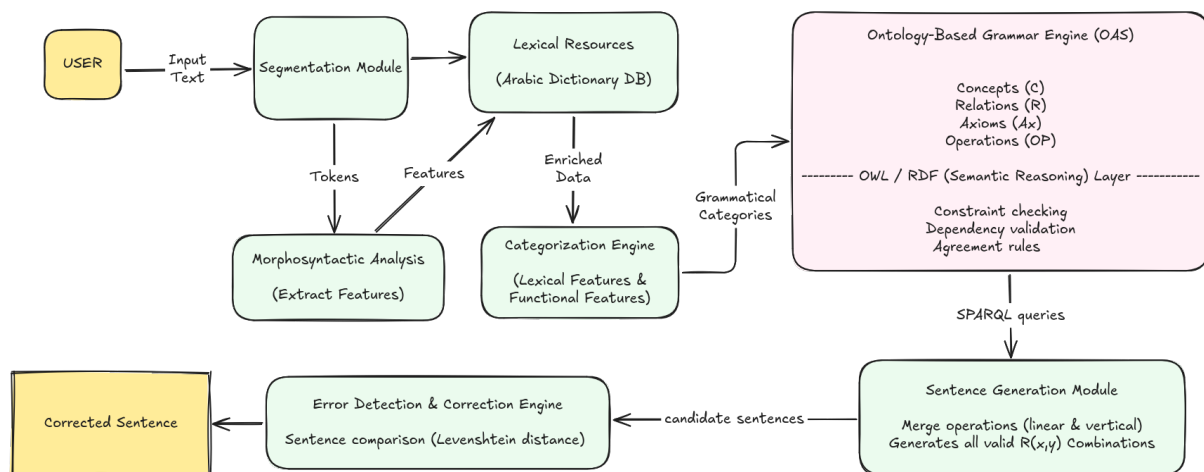
This approach leverages **Semantic Web** technologies. The semantic web operates on layers, primarily the Resource Description Framework (RDF), which represents data as subject-predicate-object triples. For Arabic grammar, the **Ontology of Arabic Syntax (OAS)** is modeled using **OWL** (Web Ontology Language), an extension of RDF that allows for highly expressive logic and reasoning. In the OAS, concepts represent grammatical categories (e.g., Noun, Verb) while relationships represent syntactic links (e.g., Subject-Verb dependency) and functional properties (e.g., Case, Gender).

The methodology employed is “Generation then Comparison”:

1. **Categorization:** Words are assigned Lexical Features (meaning-heavy aspects like Transitivity) and Functional Features (structural aspects like Number, Gender, Case).
2. **Merger (Generation):** Using **SPARQL** (the query language for RDF), the system merges words into oriented grammatical pairs. By querying the OAS, the system generates all syntactically correct combinations of the sentence that abide by Arabic grammar axioms.
3. **Comparison:** The original incorrect sentence is compared against the generated valid sentences using string distance metrics, allowing the system to deduce the required correction.



## Modular Decomposition



**Figure 4.12:** Architecture of the GEC Module.

The Ontology Engine orchestrates the generation logic via the following components:

- **Feature Mapper:** Translates CAMEL morphological outputs into the specific OWL terminology required by the OAS.
- **SPARQL Query Generator:** Constructs dynamic semantic queries. For example, it can query the ontology to enforce the axiom that a Verb's Subject must possess a Nominative case.
- **Candidate Generator:** Combines the theoretical rules from the ontology with the Morphological Generator from the Dictionary to produce the final corrected string phrase.
- **Explanation Generator:** Utilizes the specific violated ontology axiom to dynamically construct an educational, human-readable grammatical explanation for the user.

## Design Constraints

Generating every possible syntactically correct permutation of a full sentence is computationally prohibitive, posing a significant latency risk for a real-time system. To circumvent this, the generation phase is aggressively localized. Instead of processing the entire sentence, the Ontology Engine restricts its generation only to the precise error spans previously identified by the Grammatical Error Detection (GED) module. Furthermore, bridging the terminology gap between the morphological analyzer's tags and the OAS concepts required building a rigid, robust mapping dictionary to prevent query failures.

## Results



Figure 4.13: Grammatical Error (Original)



Figure 4.14: Ontology Correction

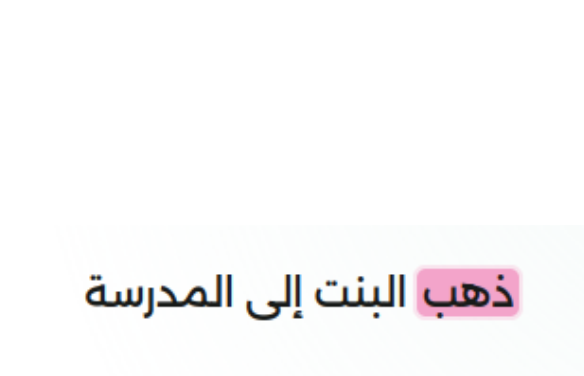


Figure 4.15: Syntactic Mismatch (Original)



Figure 4.16: Ontology Correction

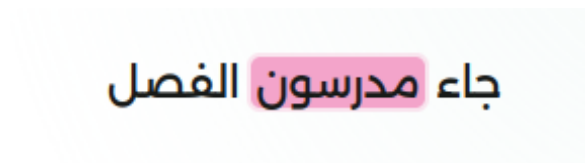


Figure 4.17: Case/Gender Error (Original)

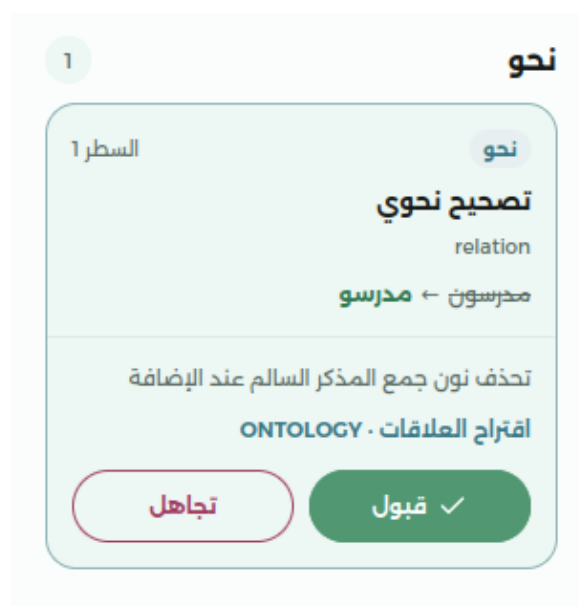


Figure 4.18: Ontology Correction

### 4.4.3. Text Editing Sub-module

#### Functional Overview

The system implements a Grammatical Error Correction (GEC) model for Arabic using a **text editing (sequence tagging)** approach rather than a traditional sequence-to-sequence (Seq2Seq) generative model. By treating correction as a sequence of edit operations rather than generating text from scratch.

#### Correction Generation

Instead of generating the corrected sentence autoregressively, the model predicts a sequence of edit operations for each token in the erroneous input. These predicted edits are then applied to the original text to produce the corrected sentence.

- **Edit Extraction & Alignment:** The system aligns erroneous and corrected sentence pairs using a weighted Levenshtein edit distance, accompanied by an iterative algorithm to capture multi-word errors. It then applies the character-level alignments.
- **Applying Operations:** It assigns specific character-level edits to the input words to transform them. The primary operations include:
  - Keep (K)
  - Delete (D)
  - Insert (I\_[])

- **Replace (R\_[])**
- **Optimization Strategies:** To keep the vocabulary of possible edits manageable and improve learning, the system uses:
  - **Edit Compression:** Merging consecutive operations (e.g., consecutive deletes DDD into D\* or insertions into I\_[ ] I\_[ \*]).
  - **Sub-word Projection:** Edit operations are first extracted at the word level, where grammatical corrections are naturally defined. These edits are then projected onto the corresponding sub-word tokens so that each token receives a label, matching the tokenization scheme used by Transformer models such as BERT.
  - **Edit Segregation:** Splitting the correction into a two-step pipeline—one model for non-punctuation errors, and a subsequent model for punctuation errors.
  - **Edit Pruning:** Removing rare edit operations during training to help the model focus on frequent and informative edits.

### Explanation Generation (Interpretability)

Unlike autoregressive Seq2Seq models, which generate only the final corrected sentence, edit-based models explicitly predict the edit operation applied to each input token (e.g., **KEEP**, **DELETE**, **REPLACE**, or **INSERT**). These edit labels provide a transparent record of how the input was transformed, making it straightforward to identify the exact grammatical changes performed. This improves the interpretability of the correction process and facilitates error analysis, debugging, and the generation of user-friendly explanations for the applied corrections.

### Modular Decomposition

- **Alignment Module**

The aligner is responsible for aligning the erroneous sentence with its corrected one using the Levenshtein edit distance algorithm. The alignment is performed in two stages. First, alignment is carried out at the **word level** to determine the correspondence between words in the two sentences. At this stage, operations such as **insertions**, **deletions**, **merges**, and **splits** are identified to handle cases where a single word in one sentence corresponds to multiple words in the other, or vice versa. Once the word-level alignment has been established, each aligned word pair is further aligned at the **character level** to determine the exact character insertions, deletions, and replacements required to transform the

erroneous word into its corrected form. This hierarchical alignment provides the detailed edit information that serves as the basis for extracting edit labels used during model training.

- **Rewrite Module**

After the initial alignment, the rewriter generates an intermediate representation of the aligned sentences. This step resolves ambiguities in the character-level alignment and rewrites complex edits into a consistent form that can be more easily projected onto tokens.

- **Edit Projection Module**

Since grammatical edits are initially extracted at the character level while Transformer-based models operate on tokenized input, the projector maps the character-level edit operations onto the corresponding tokens or sub-word tokens. Each token is assigned an edit label describing the operation that should be applied, making the extracted edits compatible with the model's token-level prediction mechanism.

- **Edit Compression Module**

The projector may produce multiple consecutive edit operations for a single token. The compressor combines these individual operations into a compact edit label whenever possible. This reduces redundancy in the label sequence, simplifies the classification problem, and produces a more manageable set of edit tags for model training and inference.

- **Edit Pruning Module**

The pruner analyses the frequency of all generated edit labels across the training corpus and removes edit labels that occur below a predefined frequency threshold. Rare edit operations contribute little to the learning process while significantly increasing the number of output classes. Pruning these infrequent labels reduces label sparsity, decreases vocabulary size, and allows the model to focus on learning the most common and informative correction patterns.

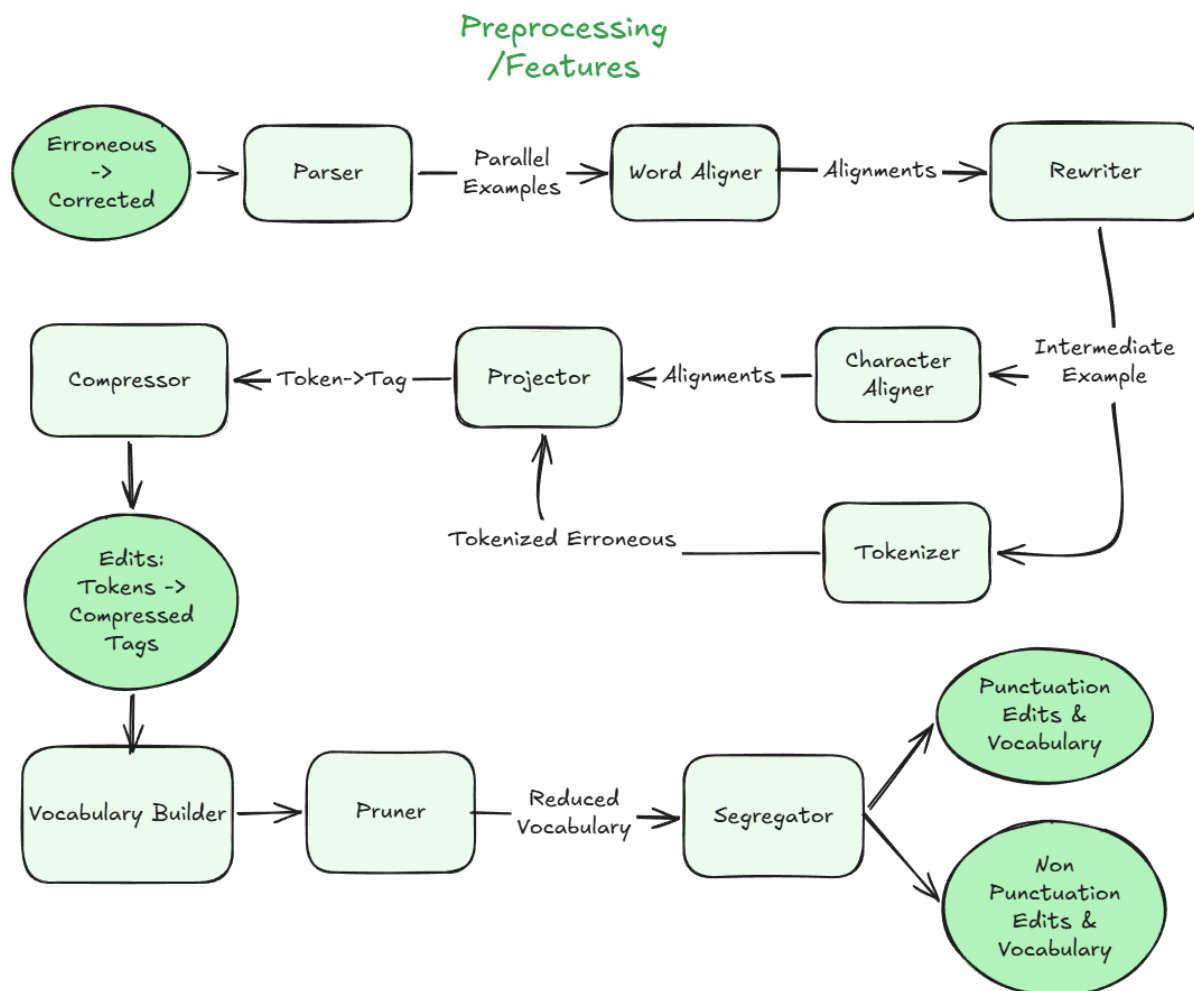
- **Edit Segregation Module**

Punctuation corrections often exhibit different characteristics from grammatical and spelling corrections. The segregator separates punctuation-related edit labels from non-punctuation edits, allowing them to be processed independently. This separation improves the handling of punctuation corrections and enables the system to treat punctuation edits differently during training or inference if required.

- **Edit Application Module**

During inference, the model predicts a sequence of edit labels rather than generating corrected text directly. The edits applicator interprets these predicted

labels and applies the corresponding operations—such as insertions, deletions, replacements, merges, or splits—to the original input tokens. The resulting sequence of modified tokens is then reconstructed to produce the final corrected sentence.



**Figure 4.19:** Tagger Pre-processing Pipeline

## Design Constraints

- The edit tagging task exhibits a significant class imbalance, as the majority of tokens in the training corpus do not require correction and are therefore assigned the KEEP label. This imbalance may bias the model toward predicting KEEP more frequently than edit operations, potentially reducing its ability to detect less common grammatical errors.
- The frequency of grammatical error types in the training corpus is highly uneven. Errors related to Hamza (إ, ء, ؤ), Taa' Marbuta (ة, ð), and punctuation constitute a large proportion of the annotated corrections, whereas many other grammatical

phenomena occur much less frequently

These corpus characteristics directly influence the model's learning behavior. Frequent edit operations are learned more reliably due to their abundance in the training data, while infrequent grammatical phenomena remain more challenging to detect and correct.

## Results



Figure 4.20

المدرسة هي ملقَى الطلاب

Figure 4.21



Figure 4.22

يجب ال إهتمام بالصحة

Figure 4.23

#### 4.4.4. Ranker Sub-module

##### Functional Description

The Ranker module is responsible for producing the final coherent text alongside its supporting grammatical explanations.

- **Correction Generation** evaluates the corrections coming from GEC making use of the GED output. It applies a rigorous filtering and scoring pipeline to candidates from multiple modules. Overlapping or conflicting candidate edits are resolved by selecting the candidate with the highest composite score.
- **Explanation Generation** is facilitated by hardcoded scoring biases that favour explainable, rule-based corrections. Specifically, if a detected error is marked as `explanation_eligible` and the candidate correction comes from the rule-based ONTOLOGY module, a positive weight (`W_EXPLAIN: +0.05`) is applied. This bias guarantees that the final selected corrections can be mapped back to transparent linguistic rules rather than relying solely on black-box predictions.

##### Modular Decomposition

This module is decomposed into four primary submodules:

1. **Configuration Manager**

Manages the tunable hyper-parameters used during scoring. Key configurations include base module weights (`W_ONTOLOGY: 0.25`, `W_TAG: 0.15`, `W_DICTIONARY: 0.10`), confidence weightings (`W_EDIT_CONF: 0.30`), and penalty weights (`W_CHAR_DIST: 0.10`, `W_LOW_CONF: 0.15`).

2. **Candidate Aggregator & Matcher**

Maps candidate edits to specific errors detected by the GED by checking span intersections. Candidates whose correction equals the original text are immediately filtered out.

3. **Scoring Engine**

Computes a comprehensive `score` for each candidate edit using multiple weighted heuristics:

- **Base Module & Confidence:** Adds base module weight, edit confidence (`W_EDIT_CONF`), and GED confidence (`W_GED_CONF`).
- **Provenance Tiers:** Boosts scores for `tier_1_rule_derived` (+0.15) and `tier_2_rule_supported` (+0.08) error spans.



- **Linguistic Alignment:** Awards +0.10 (`W_SPELL_DICT`) if a Dictionary module corrects an `ORTHOGRAPHY` error, and +0.10 (`W_GRAM_ONT`) if the Ontology module corrects a `SYNTAX` or `MORPHOLOGY` errors. Out-of-vocabulary (`is_oov`) tokens receive a +0.05 bonus.
- **Multi-Source Agreement:** Awards +0.20 (`W_AGREEMENT`) for each peer module that suggests the exact same correction string.

#### 4. Final Selection & Metadata Generator

Sorts the scored candidates per error using a strict tuple key: (`score`, `edit_confidence`, `module_priority`, `-length_of_correction`). `module_priority` is strictly ranked: Ontology (3) > Tag (2) > Dictionary (1). It then iterates chronologically through the text, applying the best candidate. Finally, it emits the final correction.

### Design Constraints

- **Grammatical Accuracy and Precision:**

The system penalizes edits that diverge too much from the original text. A penalty (`W_CHAR_DIST`: -0.10) is applied proportionately to the normalized Levenshtein distance, and another penalty (`W_LENGTH_RATIO`: -0.05) is applied for deviations in string length. Furthermore, if a candidate's `edit_confidence` falls below `CONF_LOW_THRESHOLD` (0.30), a flat penalty of -0.15 (`W_LOW_CONF`) is applied.

## 4.5. Module 3: Web Client

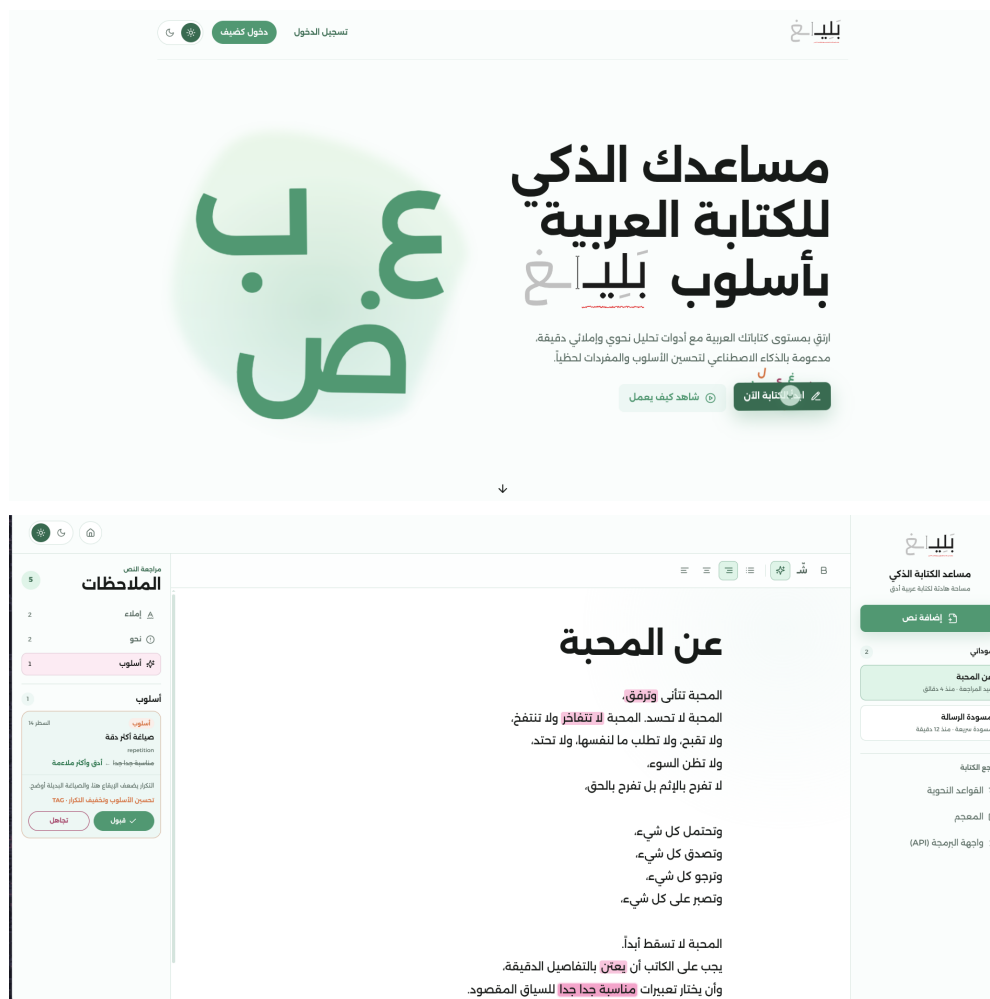
The web client is the user-facing application that exposes Baligh's capabilities through an interactive writing environment. Built with React and Vite, it provides a lightweight, design-system-aligned interface that communicates with the backend API to present analysis results, corrections, and suggestions to the user.

### 4.5.1. Functional Description

The web client serves two primary functions. First, it provides a public-facing landing experience that introduces Baligh and its capabilities. Second, it offers an editor workspace where users write Arabic text and receive real-time feedback from the GED, GEC, and NWS modules. The editor highlights detected issues inline, displays issue cards with explanations, and surfaces correction candidates and next-word suggestions through contextual UI elements.

The client is structured around reusable, design-system-aligned components that follow

an Arabic-first, learning-oriented design philosophy. The interface prioritizes clarity over visual complexity, ensuring that detector outputs appear as guided feedback rather than raw diagnostic logs.



**Figure 4.24:** The Baligh web client: the public landing page and the editor workspace used to present analysis results to the user.

# Chapter 5: System Testing and Verification

To ensure that Baligh operates correctly, achieves high linguistic accuracy, and maintains responsive real-time performance, a multi-layered testing and verification strategy was implemented. The validation framework includes automatic unit testing of isolated backend algorithms, end-to-end automated browser testing of frontend interactive elements, and evaluation of core Natural Language Processing (NLP) models on gold-standard Arabic corpora. This chapter details the testing environment setup, the modular and integrated testing methodologies, the continuous integration schedule, and the quantitative evaluation results of the grammatical error detection and next-word prediction modules.

## 5.1. Testing Setup

Testing and evaluation were conducted in a standardized environment to ensure the reproducibility of all results. The hardware and software specifications of the testing environment are defined as follows:

- **Operating System:** Linux (Ubuntu 22.04 LTS / Debian-based distributions).
- **Python Environment:** Python 3.11.x, managed using the `uv` package installer and virtual environment manager to guarantee deterministic dependency resolution.
- **Node.js Environment:** Node.js v24.x, utilizing the `pnpm` package manager (version 10.33.4) for frontend dependency caching and management.
- **Linguistic Libraries and Resources:**
  - `camel-tools` (version 1.5.7+) loaded with the Modern Standard Arabic morphology database (`morphology-db-msa-r13`) and the Maximum Likelihood Estimation disambiguator (`disambig-mle-calima-msa-r13`).
  - `farasapy` (version 0.1.1) for initial text segmentations.
- **Testing Libraries:** `pytest` (version 8.3.5) for Python backend testing; standard Node testing libraries for frontend unit tests; and `playwright` for automated end-to-end browser execution.

- **Evaluation Corpora:**

- **Qatar Arabic Language Bank (QALB):** The QALB-2014 and QALB-2015 test datasets, which contain native and non-native Arabic comments extracted from Aljazeera articles, characterized by a high frequency of spelling and punctuation errors.
- **ZAEBUC:** A bilingual Arabic-English student writing corpus used to evaluate detection and correction robustness in educational settings.
- **NWP Corpus:** An Arabic text corpus combining Arabic Wikipedia articles and the MAD corpus (totaling 900,000 test-split words) for next-word prediction models.

## 5.2. Testing Plan and Strategy

The verification strategy relies on testing components in isolation (module testing) before testing their interactions in a fully assembled pipeline (integration testing).

### 5.2.1. Module Testing

Each subsystem in Baligh is validated independently using automated unit tests to guarantee functional correctness under diverse inputs.

- **Preprocessing Service:** Unit tests in the `tests/services/preprocessing/` directory verify normalizer rules (such as Alif and Ya normalization), sentence boundary detection, word segmenters, and morphological analyzers. For example, `test_analyzer.py` validates that out-of-vocabulary (OOV) tokens (e.g., typos like “الطلاب” containing an extra letter) are correctly flagged with an `is_oov = True` attribute to prevent cascading errors in downstream grammar checks.
- **Grammatical Error Detection (GED):**
  - *Rule-Based Subsystem:* Unit tests verify syntactic, semantic, punctuation, and orthographical rules. Files like `test_syntax.py` and `test_punctuation.py` check specific grammar patterns, such as subject-verb agreement or missing spaces before punctuation.
  - *Lexicon Subsystem:* Verified in `test_detector.py` and `test_trie_runtime.py` to ensure that dictionary words and entity phrases are loaded correctly into MARISA trie data structures for rapid OOV detection.
  - *Machine Learning Subsystem:* Verified in `test_features.py` and `test_detector.py` to ensure sequence labeling features are extracted and parsed correctly

by the CRF model.

- *Fusion Layer*: Tested in `test_fusion.py` to ensure overlap resolution works according to defined tier priorities (ex., Tier 1 rules override Tier 3 statistical detections; overlapping spans of the same category are merged; duplicate sources are consolidated).
- **Grammatical Error Correction (GEC)**: Evaluated in `tests/services/gec/`.
  - *Ontology Subsystem*: Tests morphological generation rules and syntactic corrections (such as gender and case endings) by querying semantic reasoning rules.
  - *Dictionary Subsystem*: Tests spelling candidate suggestion engines, edit-distance generators, and phonetic ranking matchers.
- **Next-Word Prediction (NWP)**: Verified in `tests/services/nws/`.
  - *Next-Word Prediction (NWP) Subsystem*: Tests candidate word predictions generated from the LSTM model and the Word N-Gram model, validating their hybrid confidence blending.
  - *Word Auto-Completion (WAC) Subsystem*: Tests typing completions using the Character N-Gram model to verify MRR and KSR metrics.
  - *Prototyping & Orchestration*: Utilized interactive Jupyter Notebooks to visually evaluate beam search rankings and maintained a local orchestrator test (`test_orchestrator.py`), which was excluded from final submission to avoid requiring large CUDA library dependencies.
- **Frontend Application**: Frontend unit tests ensure utility functions, schema parsers, and editor states function correctly under standard input conditions.

### 5.2.2. Integration Testing

Integration testing validates that data flows correctly across the entire application stack.

- **GED Orchestrator Integration**: Tested in `test_ged_orchestrator.py`. This ensures that a raw Arabic sentence is properly normalized, segmented, analyzed morphologically, routed through all three detectors, and fused into a final list of resolved error spans.
- **Client-Server API Integration**: Verified by testing backend endpoints to ensure HTTP requests correctly accept Arabic payloads, apply the processing pipeline, and return JSON responses conforming to the API schemas.

- **End-to-End Browser Automation:** Using Playwright (`pnpm test:e2e`), automated Chromium browser instances simulate actual typing sessions. The E2E suite verifies that when a user types text in the editor, error spans are visually highlighted, suggestion dropdowns populate with correction candidates, and explanatory text bubbles appear beneath active cursors.

## 5.3. Test Schedule

Testing is integrated into the daily development workflow through three distinct stages:

1. **Local Pre-Commit Verification:** Code formatting (`ruff format`), linting checks (`ruff check`), and static type checking (`mypy`) are run on local changes prior to committing.
2. **Local Pre-Push Validation:** Developers run the complete check suite locally using a unified command (`make all`), which triggers formatting, linting, type-checking, and the full backend `pytest` test suite.
3. **Continuous Integration (CI):** Managed via GitHub Actions on every pull request and push to the `main` branch. The CI workflow splits verification into two parallel jobs:
  - `quality`: Sets up a Python 3.11 environment, installs `astral uv`, restores cached NLP datasets (Farasa and CAMEL Tools), overrides PyTorch with a CPU-only version to reduce runner overhead, downloads morphology databases, and executes formatting checks, linting, type checking, and unit tests.
  - `frontend`: Sets up Node.js 24 and `pnpm`, checks formatting and linting, runs frontend unit tests, installs Playwright dependencies, and executes end-to-end browser tests.

## 5.4. Comparative Results to Previous Work

The performance of Baligh's primary modules was quantitatively evaluated on established benchmarks.

### 5.4.1. Grammatical Error Detection (GED) Performance

The GED system was evaluated on QALB-2014, QALB-2015 (separate native L1 and non-native L2 sets), and ZAEBUC test sets. Performance is measured using Precision (P), Recall (R), F1-Score, and False Positives per 1000 tokens (FP/1000). The results are summarized in Table 5.1.

**Table 5.1:** Grammatical Error Detection Subsystem Performance

| Dataset             | System / Detector         | Precision (P) | Recall (R)    | F1-Score      | FP/1000      |
|---------------------|---------------------------|---------------|---------------|---------------|--------------|
| <b>QALB-2014</b>    | Rule-Based Detector       | 69.22%        | 18.76%        | 29.52%        | 18.88        |
|                     | Lexicon Matcher           | 89.64%        | 13.83%        | 23.96%        | 3.62         |
|                     | Sequence Labeler (ML)     | 92.91%        | 77.49%        | 84.50%        | 13.39        |
|                     | <b>Fused Orchestrator</b> | <b>92.03%</b> | <b>79.75%</b> | <b>85.45%</b> | <b>15.63</b> |
| <b>QALB-2015 L1</b> | Rule-Based Detector       | 71.61%        | 20.34%        | 31.68%        | 17.38        |
|                     | Lexicon Matcher           | 88.21%        | 14.97%        | 25.59%        | 4.31         |
|                     | Sequence Labeler (ML)     | 92.47%        | 81.84%        | 86.83%        | 14.37        |
|                     | <b>Fused Orchestrator</b> | <b>91.52%</b> | <b>84.30%</b> | <b>87.76%</b> | <b>16.84</b> |
| <b>QALB-2015 L2</b> | Rule-Based Detector       | 55.79%        | 7.59%         | 13.36%        | 13.57        |
|                     | Lexicon Matcher           | 88.87%        | 9.48%         | 17.13%        | 2.68         |
|                     | Sequence Labeler (ML)     | 82.67%        | 40.66%        | 54.51%        | 19.24        |
|                     | <b>Fused Orchestrator</b> | <b>82.49%</b> | <b>42.74%</b> | <b>56.31%</b> | <b>20.47</b> |
| <b>ZAEBUC</b>       | Rule-Based Detector       | 62.58%        | 19.93%        | 30.23%        | 33.41        |
|                     | Lexicon Matcher           | 96.97%        | 8.92%         | 16.34%        | 0.78         |
|                     | Sequence Labeler (ML)     | 87.63%        | 77.98%        | 82.52%        | 30.87        |
|                     | <b>Fused Orchestrator</b> | <b>87.58%</b> | <b>79.09%</b> | <b>83.12%</b> | <b>31.46</b> |
| <b>Aggregate</b>    | Rule-Based Detector       | 68.48%        | 17.39%        | 27.73%        | 17.95        |
|                     | Lexicon Matcher           | 89.17%        | 13.22%        | 23.02%        | 3.60         |
|                     | Sequence Labeler (ML)     | 91.29%        | 72.48%        | 80.81%        | 15.51        |
|                     | <b>Fused Orchestrator</b> | <b>90.50%</b> | <b>74.73%</b> | <b>81.86%</b> | <b>17.59</b> |



The empirical results show that while the rule-based and lexicon detectors exhibit high precision (especially the lexicon matcher on ZAEBUC at 96.97%), they suffer from low recall. The machine-learning sequence labeler provides the highest recall, and the fused orchestrator successfully integrates all subsystems to achieve the best overall F1-score across all test sets, raising the aggregate F1-score to 81.86%.

5.4.2. Next-Word Prediction (NWP) Performance

The Next-Word Prediction (NWP) models were evaluated on the test split of the LSTM Corpus. We compare the standalone LSTM model against a hybrid predictor that blends LSTM and Word N-Gram models (with confidence weight  $\alpha = 0.75$ ). The evaluation metrics include Top-1, Top-3, and Top-5 prediction accuracy, as shown in Table 5.2.

Table 5.2: Next-Word Prediction (NWP) Accuracy Comparison

| Model            | Top-1 Accuracy | Top-3 Accuracy | Top-5 Accuracy |
|------------------|----------------|----------------|----------------|
| Standalone LSTM  | 10.88%         | 17.56%         | 21.52%         |
| Hybrid Predictor | <b>11.40%</b>  | <b>18.20%</b>  | <b>23.00%</b>  |

The hybrid predictor outperforms the standalone LSTM model by leveraging high-frequency local word patterns from the N-Gram model alongside the long-range semantic representation of the LSTM.

5.4.3. Word Auto-Completion (WAC) Performance

The context-aware Character N-Gram model for Word Auto-Completion was evaluated under typing scenarios. It achieves a Top-1 accuracy of 30.00%, Top-3 accuracy of 40.80%, and Top-5 accuracy of 46.40%, with a Mean Reciprocal Rank (MRR) of 0.3699 and a Keystroke Savings Rate (KSR) of 7.40%. The performance improves significantly as the length of the typed prefix increases, as detailed in Table 5.3.

Table 5.3: WAC Top-5 Accuracy by Typed Prefix Length

| Prefix Length (chars) | 1      | 2      | 3      | 4      | 5      | 6+      |
|-----------------------|--------|--------|--------|--------|--------|---------|
| Top-5 Accuracy        | 11.76% | 32.79% | 54.00% | 77.50% | 94.44% | 100.00% |

As the user types, the character prefixes restrict the possible lexical search space, causing the top-5 prediction accuracy to rise from 11.76% for a single character to 100.00% once 6 characters have been entered.

## Chapter 6: Conclusions and Future Work

This chapter summarizes the overall journey of developing the Baligh project. It details the various challenges encountered during the research, design, and implementation phases, along with the strategies used to overcome them. Furthermore, it outlines the valuable experiences and technical skills gained by the team members. Finally, the chapter presents the overarching conclusions drawn from the project, highlighting its main features and limitations, and provides strategic directions for future work to expand and improve upon the current system.

### 6.1. Faced Challenges

Throughout the development of Baligh, the team encountered several research, technical, and integration challenges. To ensure a modular architecture and facilitate future updates, these challenges are categorized by the respective system modules.

#### 6.1.1. Research Challenges

- **Next-Word Suggestion (NWS) Module:** One of the primary difficulties encountered during the research phase was the poorness of literature on Next-Word Suggestion for the Arabic language. The available research was found to be limited in quantity and often lacking in quality. To compensate for this gap and achieve reliable results, it was necessary to look beyond Arabic and review papers focusing on other morphologically rich languages that share similar structural complexities.

#### 6.1.2. Technical and Integration Challenges

- **Next-Word Suggestion (NWS) Module:** Developing a real-time predictive text module posed significant challenges, primarily revolving around the trade-off between latency and accuracy, data scarcity, and poorly existing literature.
  - **Model Latency vs. Accuracy:** Balancing the fast response time required for typing with the context-awareness of neural models was difficult. This was done by implementing a hybrid approach that blends a fast Word N-Gram model with an LSTM model.

- **Misleading Literature on N-Gram Performance:** Most papers claiming that simple N-Gram models are highly fit for NWS tasks often exaggerate their performance and use unrealistic numbers, which caused us to question our own results for a time. Furthermore, these studies assume the availability of massive corpora containing hundreds of millions to over a billion tokens to achieve good accuracy. Due to computing power limitations, training on such massive datasets was not feasible for our project.
- **Data Scarcity for Interactive Evaluation:** While there are standard corpora for Arabic, finding datasets that closely mimic user keystrokes for evaluating Word Auto-Completion was challenging. We had to synthesize evaluation strategies using MRR and KSR metrics.
- **Grammatical Error Detection (GED) Module**
  - **High False Positives in Statistical Models:** Initial sequence labeling models often over-flagged correct Arabic structures. This was addressed by introducing a fusion orchestrator that prioritizes rule-based and lexicon-based detections over statistical models.
  - **Morphological Ambiguity:** Arabic's rich morphology often caused the system to misidentify word boundaries or grammatical roles. Integrating Farasa and CAMEL Tools for accurate morphological analysis prior to detection was crucial to resolving this.
- **Grammatical Error Correction (GEC) Module**
  - **Enormous Search Space:** Generating correction candidates for out-of-vocabulary words yielded too many possibilities. We overcame this by using dictionary lookups paired with phonetic and edit-distance ranking to surface only the most relevant suggestions.
  - **Complex Rule Maintenance:** Building an ontology-based correction module required extensive manual encoding of Arabic grammatical rules. Keeping this rule set manageable and efficient required careful software design.
- **Integration and Web Application**
  - **Pipeline Latency:** Running GED, GEC, and NWS sequentially for every user keystroke caused noticeable delays. We resolved this by optimizing the MongoDB client, separating analysis onto background threads, and making NWS requests independent from grammar checks.
  - **Rich Text Editor Alignment:** Ensuring that the highlight spans from the backend exactly matched the user's text on the frontend—especially after

whitespace modifications or partial edits—required precise index mapping and synchronization.

## 6.2. Gained Experience

Working on Baligh provided the team with extensive hands-on experience in both technical and non-technical domains. The collaborative nature of the project allowed team members to specialize while maintaining a unified system.

### 6.2.1. Technical Skills

- **Natural Language Processing (NLP):** The team gained profound insights into the linguistic differences of Arabic NLP. This included developing robust pipelines for morphological analysis, tokenization, and handling orthographic ambiguities. We also gained practical experience in extracting meaningful features for sequence labeling algorithms like CRFs and applying N-Gram structures effectively for predictive tasks.
- **Machine Learning (ML) & Data Science:** Extensive experience was gained in the entire ML lifecycle, from curating and preprocessing messy Arabic text corpora to training, evaluating, and fine-tuning models. We learned to balance accuracy and latency by evaluating different architectures, such as blending LSTMs with N-Grams, and measuring performance using metrics like MRR and KSR.
- **Frontend UI Design and Backend Integration:** Although heavily streamlined during development, constructing the interactive interface strengthened our skills in custom, handcrafted UI design. Setting up the core backend also provided practical experience in handling API integrations and orchestrating requests efficiently.

### 6.2.2. Soft Skills and Teamwork

- **Modular Collaboration:** Dividing the project into NWS, GED, GEC, and Integration taught the team how to define clear API contracts and interfaces, allowing parallel development without blocking one another.
- **Version Control and CI/CD:** Utilizing Git, GitHub Actions, and rigorous code review processes ensured code quality and seamless integration of independently developed modules.
- **Problem Solving and Research:** The project required translating academic re-

search into practical engineering solutions, bridging the gap between theoretical Arabic linguistics and applied software engineering.

## 6.3. Conclusions

The Baligh project successfully demonstrates that a comprehensive, real-time Arabic writing assistant is achievable by combining traditional linguistic rules with statistical and machine learning approaches. The final system integrates Grammatical Error Detection, Grammatical Error Correction, and Next-Word Suggestion to provide a better user experience.

### **Key Features achieved by the project include:**

- A hybrid detection and correction pipeline that accurately addresses a wide range of Arabic grammatical and orthographic errors.
- A fast, context-aware next-word prediction and auto-completion engine that reduces typing effort.
- Educational, explainable feedback that helps users understand their mistakes rather than just blindly accepting corrections.
- A highly responsive web interface customized for the Arabic language.

### **Current Limitations of the system include:**

- The reliance on word-level and character-level models in the NWS module, which can occasionally miss deep semantic context.
- The ontology and rule-based corrections, while explainable, are currently limited in coverage and may not handle rare or complex grammatical structures.
- The system is primarily optimized for Modern Standard Arabic (MSA) and may struggle with dialectal Arabic.

## 6.4. Future Work

To build upon the foundation established by Baligh, subsequent development can focus on scaling the system and incorporating more advanced technologies.

### 6.4.1. Next-Word Suggestion (NWS) Enhancements

- **In-Sentence Completions:** Expanding the suggestion capabilities to support completions anywhere within the sentence context, rather than strictly predicting the next word at the end of a sentence.
- **Extensive Model Benchmarking:** Experimenting with and benchmarking a wider variety of models to determine the highest achievable results and accuracies for Arabic text.
- **Domain-Specific Training and User Personalization:** Training models on higher volume datasets and providing distinct models customized for specific text domains (e.g., legal, medical, or casual text). The system could then dynamically switch to a model based on the specific domain of the user currently using the application.

### 6.4.2. Grammatical Error Detection and Correction Enhancements

- **End-to-End Seq2Seq Models:** Integrating an advanced sequence-to-sequence model alongside the text-editing approach to handle complex, multi-word structural corrections.
- **Expanded Ontology:** Broadening the rule-based ontology to cover deeper semantic errors and stylistic improvements, beyond basic syntax and orthography.
- **Contextual Spelling Correction:** Utilizing contextual embeddings (like AraBERT) to detect real-word spelling errors where the word is correctly spelled but incorrect in context.

### 6.4.3. System and Platform Extensions

- **Dialectal Support:** Expanding the datasets and models to support common Arabic dialects alongside MSA.
- **Browser Extension & Plugins:** Developing browser extensions and plugins for popular word processors (e.g., Microsoft Word, Google Docs) to make Baligh accessible across different platforms.
- **Offline Mode:** Packaging the core lightweight models (like N-Grams and MARISA Tries) for offline usage on mobile devices or desktop applications.

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# Chapter A: Development Platforms and Tools

This appendix details the comprehensive technology stack, hardware, and datasets utilized throughout the development, testing, and deployment phases of the Baligh system.

## Programming Languages

- **Python ( $\geq 3.11$ ):** The primary language for the core NLP backend, machine learning inference, and API orchestration.
- **TypeScript & JavaScript:** Used for developing the web frontend application and the browser extension.
- **HTML5 & CSS3:** Core web technologies structuring and styling the user interfaces.

## Backend Frameworks and Libraries

- **FastAPI & Uvicorn:** High-performance ASGI framework and web server used for serving the RESTful API endpoints.
- **PyTorch ( $\geq 2.4$ ):** The deep learning framework powering the neural network models.
- **HuggingFace Transformers & Accelerate:** Utilized for loading, fine-tuning, and running transformer-based language models efficiently.
- **Scikit-Learn & sklearn-crfsuite:** Employed for traditional machine learning approaches and Conditional Random Field (CRF) sequence tagging.
- **CAMEL Tools & FarasaPy:** Specialized Arabic Natural Language Processing libraries used extensively for text preprocessing, tokenization, and morphological analysis.
- **Pydantic:** Used for strict data validation and serialization schemas across the API.

- **Motor:** The asynchronous Python driver utilized for non-blocking communication with MongoDB.
- **SPARQLWrapper:** A Python wrapper around SPARQL services used to interface with the Ontology of Arabic Syntax (OAS).
- **Marisa-Trie:** A static and memory-efficient Trie data structure library used for fast lexical lookups.

## Frontend Frameworks and Libraries

- **React 19 & React Router DOM:** The core UI library and routing solution for building the interactive single-page application.
- **Vite:** A fast, modern build tool and development server for the frontend workspace.
- **TailwindCSS v4:** A utility-first CSS framework enabling rapid, responsive styling.
- **React Aria Components & Framer Motion:** Used to construct accessible, highly interactive, and smoothly animated UI components.
- **Lucide React:** Providing scalable vector iconography throughout the application interfaces.

## Databases and Storage

- **MongoDB:** A NoSQL document database used for scalable data persistence (e.g., storing user profiles, document histories).
- **SQLite:** A lightweight, file-based relational database employed locally to house the Arramooz lexical dictionary and ontology relational data for rapid local queries.

## Datasets

- **Qatar Arabic Language Bank (QALB):** The QALB-2014 and QALB-2015 test datasets served as the primary benchmark corpora. These datasets contain native and non-native Arabic comments extracted from Aljazeera articles and are characterized by a high frequency of authentic spelling and punctuation errors.
- **Zayed Arabic English Bilingual Undergraduate Corpus (ZAEBUC):** An annotated corpus containing essays written by university students, utilized to evaluate models on authentic learner errors.

- **Arabic Wikipedia:** Data extracted from Arabic Wikipedia was used for large-scale language modeling, vocabulary extraction, and providing general-domain context for the predictive models.

## Development and CI/CD Tools

- **Version Control & Automation:** Git, GitHub, and GitHub Actions formed the backbone of the continuous integration and delivery pipelines.
- **Package Management:** `uv` was utilized for extremely fast Python dependency management and virtual environments; `pnpm` managed Node.js packages in the frontend monorepo.
- **Backend QA:** **Ruff** (linting and formatting), **MyPy** (static type checking), and **Pytest** (unit testing framework).
- **Frontend QA:** **ESLint & Prettier** (linting and formatting), **Vitest** (unit testing), and **Playwright** (end-to-end browser testing).
- **Task Automation:** **GNU Make** (via `Makefile`) and **pre-commit** hooks were used to orchestrate development workflows and enforce code quality standards before commits.

## Chapter B: Use Cases

The Baligh system serves a diverse set of users by addressing their specific Arabic writing needs:

- **Arabic Language Learners:** To understand their grammatical mistakes through explainable corrections and improve their writing skills.
- **Content Creators & Journalists:** To quickly draft, proofread, and ensure their articles are free from linguistic errors before publication.
- **Teachers & Educators:** To assist in grading essays or preparing grammatically sound educational materials.
- **Lawyers & Professionals:** To draft formal documents, contracts, and emails with high linguistic accuracy and professionalism.
- **Companies & Institutions:** To maintain standardized, error-free Arabic communication across their platforms and internal tools.
- **Translators:** To ensure that content translated into Arabic is grammatically sound, flows naturally, and uses appropriate vocabulary.

## Chapter C: User Guide

For a demonstration of how to use the Baligh system, please refer to our video demo:

[https://www.youtube.com/watch?v=1NjsZ\\_7A3k8](https://www.youtube.com/watch?v=1NjsZ_7A3k8)

Additional details, including step-by-step instructions on how to install and run the project, can be found in the project's README repository: <https://github.com/El-Maktab/baligh>

## Chapter D: Code Documentation

*Document important modules, classes, APIs, configuration files, and code structure.*

## Chapter E: Feasibility Study

*This appendix should be included. Add market, technical, operational, and financial feasibility details.*

## Chapter F: GED Rules Reference

This appendix provides a comprehensive reference of the rule inventory that supports the implemented GED subsystem. The rules are organized by category: orthography, punctuation, syntax, and semantics. The lexicon split and merge patterns are listed separately. The rule-based catalog is maintained in the project documentation, while the split and merge patterns are defined in the lexicon detector resource files.

### F.1. Rule-Based GED Catalog

#### F.1.1. Orthography Rules

| Rule ID                 | Description                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| OT_HAMZA_PREP           | حروف الجر والربط وبعض الأدوات التي أصلها بهمزة قطع تكتب بالهمزة لا بالألف المجردة، مثل: إلى، أو، إذا، أن، إن. |
| OT_ALIF_MAQSURA_ALA     | الصواب «على» لا «علي».                                                                                        |
| OT_ALIF_MAQSURA_HATTA   | الصواب «حتى» لا «حتي».                                                                                        |
| OT_TANWIN_NASB_ON_ALIF  | تنوين النصب يكتب على الحرف الذي قبل الألف لا على الألف نفسها.                                                 |
| OT_IDGHAM_AN_MA         | الأفصح إدغام «عن ما» في «عَمَّا» عندما تليها جملة فعلية.                                                      |
| OT_IDGHAM_MIN_MA        | الأفصح إدغام «من ما» في «مَمَّا» عندما تليها جملة فعلية.                                                      |
| OT_TA_MARBUTA_NOUN      | وجوب كتابة التاء المربوطة «ة» في أواخر الأسماء المؤنثة بدل الهاء «ه».                                         |
| OT_TA_MARBUTA_ADJ       | وجوب كتابة التاء المربوطة «ة» في أواخر الصفات المؤنثة بدل الهاء «ه».                                          |
| OT_TA_MARBUTA_NOUN_PROP | وجوب كتابة التاء المربوطة «ة» في أواخر الأعلام المؤنثة بدل الهاء «ه»، مثل: مكة، فاطمة.                        |
| OT_INNA_AFTER_QAWL      | همزة «إن» تُكسر وجوباً بعد القول.                                                                             |
| OT_IDHA_HAMZA           | «إذا» تُكتب بهمزة قطع تحت الألف.                                                                              |
| OT_ARWAH_HAMZA          | كلمة «أرواح» جمع تكسير تُكتب بهمزة قطع على الألف.                                                             |
| OT_AQAL_HAMZA           | أفعل التفضيل «أقل» يُكتب بهمزة قطع على الألف.                                                                 |



## F.1.2. Punctuation Rules

| Rule ID                   | Description                                                                         |
|---------------------------|-------------------------------------------------------------------------------------|
| PC_SPACE_BEFORE_PUNC      | وجوب اتصال علامات الترقيم العربية مباشرة بالكلمة التي تسبقها دون فاصل أو مسافة.     |
| PC_LATIN_COMMA_ARABIC     | في السياق العربي تستعمل الفاصلة العربية «،» لا الفاصلة اللاتينية.                   |
| PC_LATIN_QUESTION_ARABIC  | في السياق العربي تستعمل علامة الاستفهام العربية «؟» لا العلامة اللاتينية.           |
| PC_LATIN_SEMICOLON_ARABIC | في السياق العربي تستعمل الفاصلة المنقوطة العربية «؛» لا العلامة اللاتينية المناظرة. |

## F.1.3. Syntax Rules

| Rule ID                            | Description                                                                  |
|------------------------------------|------------------------------------------------------------------------------|
| SY_DEM_HADHANI_FEM                 | الصواب «هاتان» مع الاسم المثنى المؤنث لا «هذان».                             |
| SY_DEM_HATANI_MASC                 | الصواب «هذان» مع الاسم المثنى المذكر لا «هاتان».                             |
| SY_DEM_HADHAYNI_FEM                | الصواب «هاتين» مع الاسم المثنى المؤنث لا «هذين».                             |
| SY_DEM_HATAYNI_MASC                | الصواب «هذين» مع الاسم المثنى المذكر لا «هاتين».                             |
| SY_DEM_HADHANI_CASE_OBLIQUE_NOUN   | إذا جاء بعد «هذان» اسم مثنى مذكر غير مرفوع فهناك خلل في المطابقة الإعرابية.  |
| SY_DEM_HADHAYNI_CASE_NOM_NOUN      | إذا جاء بعد «هذين» اسم مثنى مذكر مرفوع فهناك خلل في المطابقة الإعرابية.      |
| SY_DEM_PREP_HADHAYNI_CASE_NOM_NOUN | بعد حرف الجر يكون الاسم بعد «هذين» مجرورًا لا مرفوعًا.                       |
| SY_DEM_HATANI_CASE_OBLIQUE_NOUN    | إذا جاء بعد «هاتان» اسم مثنى مؤنث غير مرفوع فهناك خلل في المطابقة الإعرابية. |
| SY_DEM_HATAYNI_CASE_NOM_NOUN       | إذا جاء بعد «هاتين» اسم مثنى مؤنث مرفوع فهناك خلل في المطابقة الإعرابية.     |
| SY_DEM_PREP_HATAYNI_CASE_NOM_NOUN  | بعد حرف الجر يكون الاسم بعد «هاتين» مجرورًا لا مرفوعًا.                      |
| SY_LAM_JUSSIVE                     | الفعل المضارع بعد «لم» يجب أن يكون مجزومًا.                                  |
| SY_LAMMA_JUSSIVE                   | الفعل المضارع بعد «لما» يجب أن يكون مجزومًا.                                 |
| SY_LA_NAHIYA_JUSSIVE               | الفعل المضارع بعد «لا» الناهية يجب أن يكون مجزومًا.                          |
| SY_LA_NAFIYA_NOT_JUSSIVE           | «لا» النافية لا تجزم الفعل المضارع.                                          |

| Rule ID                         | Description                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------|
| SY_INNA_SISTERS_DUAL_ACCUSATIVE | أخوات إن تنصب الاسم، والمثنى بعدها يكون منصوباً بالياء لا مرفوعاً.                                        |
| SY_VERB_SUBJECT_VSO             | وجوب أفراد الفعل وتجريده من علامات التثنية والجمع إذا تقدم على الفاعل الظاهر في الجملة الفعلية.           |
| SY_NOUN_ADJ_DEFINITENESS        | وجوب مطابقة النعت للمنعوت في التعريف والتذكير.                                                            |
| SY_DEMONSTRATIVE_NOUN_GENDER    | وجوب مطابقة اسم الإشارة للاسم الذي بعده في التذكير والتأنيث، مثل: هذا البطل، هذه السلامة.                 |
| SY_RELATIVE_PRONOUN_GENDER      | وجوب مطابقة الاسم الموصول للاسم السابق له في التذكير والتأنيث، مثل: القول الذي، الوشاية التي.             |
| SY_PREP_DUAL_CASE               | وجوب جر الاسم المثنى بعد حرف الجر، مثل: في الكتابين لا في الكتابان.                                       |
| SY_PREP_SOUND_MASC_PLURAL_CASE  | وجوب جر جمع المذكر السالم بعد حرف الجر، مثل: مع المسافرين لا مع المسافرين.                                |
| SY_MAMNOO3_SARF_TANWEEN_NASB    | الممنوع من الصرف لا يُنَوَّن.                                                                             |
| SY_KAMA_ANNE                    | الصواب «كما أن» بدون الواو الزائدة.                                                                       |
| SY_LA_SIYYAMA_WA                | الصواب «لا سيما» دون واو بعدها.                                                                           |
| SY_AS_KA                        | استخدام الكاف للصفة استخدام أجنبي، والأصح النصب على الحالية.                                              |
| SY_LA_PAST_TENSE                | عند نفي الفعل الماضي يُنفى بـ «ما» أو «لم» مع المضارع، ولا يصح استخدام «لا» إلا إذا تكررت أو كانت للدعاء. |
| SY_MUBTADA_NAKIRA_PREP          | المبتدأ النكرة يؤخر وجوباً إذا كان الخبر شبه جملة من جار ومجرور.                                          |
| SY_QAAMA_BI                     | تعبير «القيام بـ» ركيك يستحسن تجنبه واستعمال الفعل مباشرة.                                                |
| SY_HAWLA_FI                     | يفضل استعمال حرف الجر «في» بدلاً من «حول» في هذا السياق.                                                  |
| SY_RAGHBA_LI                    | يُقال: «رغبة في» وليس «رغبة لـ».                                                                          |
| SY_AATIL_AN                     | الأفصح أن يقال: «عاطل من العمل» أو «متعطّل».                                                              |
| SY_TA2MEEN_DHIDDA               | الأفصح «تأمين من» أو «مناعة من» بدلاً من «ضد».                                                            |

#### F.1.4. Semantics Rules

| Rule ID         | Description                                                       |
|-----------------|-------------------------------------------------------------------|
| SE_DECADES_IYAT | يفضل في العقود الكتابة بصيغة «ينيات» لا «ينات»، مثل: الثلاثينيات. |
| SE_MOAKHARAN    | يفضل تجنب «مؤخراً» بهذا المعنى، والأفصح: حديثاً أو قريباً.        |
| SE_MUTAAKID     | يفضل استعمال «متحقق» أو «متيقن» بدل «متأكد».                      |

| Rule ID      | Description                                                            |
|--------------|------------------------------------------------------------------------|
| SE_DHATA     | يفضل «ذواتا» بدل «ذاتا» في هذا الاستعمال.                              |
| SE_KHATIR    | يفضل في هذا الاستعمال: شديد الخطر أو محفوف بالخطر بدل «خطير».          |
| SE_KHAMMARA  | «الخمار» بائعة الخمر، ويستحسن للمكان: «مخمرة» أو «حانة».               |
| SE_INDHAHALA | الفعل الأفصح: «ذهل» لا «انذهل».                                        |
| SE_BIAKMALIH | يفضل «كله» أو «جميعه» أو «برمته» بدل «بأكمله».                         |
| SE_TAHAMMAMA | يفضل استعمال «استحم» بدل «تحمم».                                       |
| SE_TASHKILU  | يستحسن الاستغناء عن «تشكل» في هذا الاستعمال أو استبدالها بـ«هي».       |
| SE_TASAMAMA  | الفعل المضعف هنا يدغم، والأفصح: «تصام» لا «تصامم».                     |
| SE_TATMIN    | الأفصح: «طمأنة» لا «تطمين».                                            |
| SE_TA3KISU   | يفضل في هذا السياق «تظهر» أو «تفصح» بدل «تعكس».                        |
| SE_JANOABI   | في الظرفية المكانية يفضل «جنوب» بدل «جنوبي».                           |
| SE_KHESISAN  | يفضل «خَصَّيَصَى» أو «خاصًا» بدل «خصيصًا».                             |
| SE_KHALOOQ   | يفضل في الوصف هنا: «حسن الخلق» بدل «خلق».                              |
| SE_RAGHMA    | يفضل «على الرغم» أو «بالرغم» أو «على» أو «مع» بدل «رغم».               |
| SE_RAFAH     | المستعمل هنا «رفات» لا «رفاة».                                         |
| SE_SHAWYAN   | الأفصح في مصدر «شوى»: «شَيَّا» لا «شويا».                              |
| SE_ARAYA     | يفضل في هذا المعنى «عريان» لا «عرايا».                                 |
| SE_LIWAHDIHI | الأفصح: «وحده» بدل «لوحده».                                            |
| SE_MAHALAT   | جمع «محل» في هذا الاستعمال هو «محال» لا «محلات».                       |
| SE_NAFOUKH   | الأفصح: «يافوخ» بدل «نافوخ».                                           |
| SE_NASHET    | في الوصف يفضل «نشيط» أو «ناشط» بدل «نشط».                              |
| SE_WALLATI   | يستحسن حذف الواو من «والتي» في هذا الربط.                              |
| SE_WALLADHI  | يستحسن حذف الواو من «والذي» في هذا الربط.                              |
| SE_ITTILAL3  | الأفصح: «اطلاع» بدل «إطلاع».                                           |
| SE_IDHTARADA | الأفصح: «اطرد» بدل «اضطرد».                                            |
| SE_MUJBAA    | الأفصح: «مجببة» بدل «مجابة».                                           |
| SE_MOUSOD    | الأفصح: «موصد» بدل «موصود».                                            |
| SE_MUASHIRAT | يستحسن في هذا المعنى: إشارات أو علامات أو شواهد أو دلائل بدل «مؤشرات». |

| Rule ID        | Description                                                        |
|----------------|--------------------------------------------------------------------|
| SE_MISHWAR     | يستحسن تجنب «مشوار» بهذا المعنى، والأفصح: طريق أو مسار أو نزهة.    |
| SE_MACHINE     | الأفصح: «مكنة» بدل «ماكينة».                                       |
| SE_IMKANIYAT   | يفضل «إمكانات» بدل «إمكانيات».                                     |
| SE_TAWAJUD     | يفضل في هذا الاستعمال «وجد» بدل «تواجد».                           |
| SE_MUSBAQAN    | يفضل «مقدمًا» أو «سلفًا» أو «قبلاً» بدل «مسبقًا».                  |
| SE_WABITTALI   | «وبالتالي» تعبير ركيك في هذا السياق، ويستحسن استبداله ببدايل أفصح. |
| SE_BIMATHABATI | يفضل «بمنزلة» أو «تقوم مقام» بدل «بمثابة».                         |
| SE_ISTIBYAN    | يفضل «استبانة» بدل «استبيان».                                      |
| SE_AKHISSAI    | يفضل «اختصاصي» بدل «أخصائي».                                       |
| SE_TOQOS       | يفضل «شعائر» بدل «طقوس» في هذا الاستعمال.                          |
| SE_BIHASABI    | يستحسن تجنب «بحسب» في هذا الاستعمال واستبدالها ببدايل أفصح.        |
| SE_LISALHIKA   | يفضل «لمصلحتك» بدل «لصالحك».                                       |
| SE_LISALIHI    | يفضل «لمصلحة» بدل «لصالح».                                         |
| SE_I3TABAR     | يفضل في هذا الاستعمال «عدّ» بدل «اعتبر».                           |
| SE_BURHA       | يفضل «هنيهة» بدل «برهة» في هذا السياق.                             |
| SE_AAWINA      | يفضل «أوان» بدل «آونة».                                            |
| SE_BAWASIL     | في هذا الاستعمال يفضل «بسلاء» أو «باسلون» بدل «بواسل».             |
| SE_MUSTAHTIR   | يفضل «مستهين» أو «مستخفّ» بدل «مستهتر» في هذا الاستعمال.           |
| SE_SUWAH       | يفضل «سياح» بدل «سواح».                                            |
| SE_KAKULL      | «ككل» ترجمة ركيكة، ويستحسن «كليًا».                                |

## F.2. Lexicon Split and Merge Patterns

The lexicon matcher supplements the rule-based system with exact curated patterns. The split and merge entries form an important bridge between pure spelling normalization and grammar-aware error detection.

### F.2.1. Split Patterns

| Pattern ID          | Wrong Surface Form | Correct Form | Purpose                                |
|---------------------|--------------------|--------------|----------------------------------------|
| LEX_SPLIT_LAKIN     | لا كن              | لكن          | Common split correction                |
| LEX_SPLIT_HAYTHU_MA | حيث ما             | حيثما        | Conditional/relative split correction  |
| LEX_SPLIT_KAY_LA    | كي لا              | كيلا         | Common split correction                |
| LEX_SPLIT_BAADAMA   | بعد ما             | بعدهما       | Common split correction                |
| LEX_SPLIT_LIMAN     | ل من               | لمن          | Clitic attachment correction           |
| LEX_SPLIT_FI_MA     | في ما              | فيما         | Common split correction                |
| LEX_SPLIT_LIMADHA   | ل ماذا             | لماذا        | Interrogative clitic correction        |
| LEX_SPLIT_INDA_MA   | عند ما             | عندما        | Common split correction                |
| LEX_SPLIT_KULLAMA   | كل ما              | كلما         | Conditional/time expression correction |
| LEX_SPLIT_RUBBA_MA  | رب ما              | ربما         | Common split correction                |

## F.2.2. Merge Patterns

| Pattern ID              | Wrong Token | Correct Tokens | Purpose                 |
|-------------------------|-------------|----------------|-------------------------|
| LEX_MERGE_IN_SHA_ALLAH  | انشاءالله   | إن شاء الله    | Common merge correction |
| LEX_MERGE_MASHA_ALLAH   | ماشاءالله   | ما شاء الله    | Common merge correction |
| LEX_MERGE_ALHAMD_LILLAH | الحمدلله    | الحمد لله      | Common merge correction |

| Pattern ID                | Wrong Token         | Correct Tokens         | Purpose                 |
|---------------------------|---------------------|------------------------|-------------------------|
| LEX_MERGE_SUBHAN_ALLAH    | سبحانالله           | سبحان الله             | Common merge correction |
| LEX_MERGE_LA_SIYYAMA      | لاسيما              | لا سيما                | Common merge correction |
| LEX_MERGE_JAZAK_ALLAH     | جزاكالله            | جزاك الله              | Common merge correction |
| LEX_MERGE_TAWAKKAL_ALLAH  | توكلتعلالله         | توكلت على الله         | Common merge correction |
| LEX_MERGE_ASTAGHFIR_ALLAH | استغفرالله          | أستغفر الله            | Common merge correction |
| LEX_MERGE_ALLAHU_AKBAR    | اللهأكبر            | الله أكبر              | Common merge correction |
| LEX_MERGE_BISMILLAH       | بسماللهالرحمنالرحيم | بسم الله الرحمن الرحيم | Common merge correction |